

Supplementary Material for “A Comprehensive Survey of Ensemble Clustering: Taxonomies, Methodologies, Experimental Benchmark, and Challenges”

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I. SUPPLEMENTARY ANALYSIS OF MAJOR METHOD CATEGORIES

A. Supplementary Analysis of CA Matrix Refinement

Although the CA matrix provides a simple and label-invariant representation of ensemble agreement, the raw CA values are not always reliable. Each entry only records how frequently two samples are placed in the same cluster across base partitions. This frequency may be affected by unstable base clusterings, redundant ensemble members, inappropriate parameter settings, or the dominance of large clusters. Therefore, a high value in the CA matrix does not always indicate a truly reliable relation, and a low value does not necessarily mean that two samples should be separated. This observation motivates a line of work that treats the CA matrix not as the final consensus evidence, but as an initial representation that needs to be refined.

A representative example is the CA matrix self-enhancement framework proposed by Jia et al. [1]. The method starts from the assumption that the raw CA matrix contains a small number of highly reliable pairwise relations. These relations can serve as anchors for refining the whole matrix. Specifically, high-confidence pairs are selected by a thresholding operation,

$$\Omega = \{(i, j) \mid \mathbf{A}_{ij} \geq \alpha\}, \quad \mathbf{H} = \Psi_{\Omega}(\mathbf{A}),$$

where α controls the confidence level and $\Psi_{\Omega}(\cdot)$ keeps only the entries indexed by Ω . Unlike direct thresholding methods, this operation does not simply discard weak relations and cluster the remaining graph. Instead, it constructs a sparse high-confidence structure that guides the reconstruction of an enhanced CA matrix.

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The enhanced matrix is obtained by solving

$$\min_{\mathbf{C}, \mathbf{E}} \text{tr}(\mathbf{C}^{\top} \Phi \mathbf{C}) + \frac{\lambda}{2} \|\mathbf{E}\|_F^2$$

$$\text{s.t. } \mathbf{A} = \mathbf{C} + \mathbf{E}, \Psi_{\Omega}(\mathbf{E}) = 0, \mathbf{C} = \mathbf{C}^{\top}, 0 \leq \mathbf{C} \leq 1,$$

where $\Phi = \mathbf{D} - \mathbf{H}$ is the graph Laplacian induced by the high-confidence matrix $\mathbf{H}$. This formulation has two complementary effects. The first term propagates reliable pairwise information through Laplacian regularization. If two samples are connected by high-confidence evidence, their neighborhood structures in the enhanced matrix are encouraged to be similar. In this sense, the method does not only preserve reliable entries, but also uses them to regularize nearby uncertain relations. The second term separates the original CA matrix into a refined consensus part $\mathbf{C}$ and a noise part $\mathbf{E}$. The constraint $\Psi_{\Omega}(\mathbf{E}) = 0$ prevents the model from treating selected high-confidence relations as noise, which makes these relations stable anchors during refinement.

This method is useful because it addresses a common weakness of raw CA evidence. The original CA matrix is purely accumulative. It counts co-occurrence but does not distinguish between reliable and unreliable agreement. Jia et al.’s method introduces a reliability-aware view: only the most confident relations are used as structural guidance, while other entries are adjusted under this guidance. This makes the enhanced matrix less sensitive to noisy base partitions and helps recover a smoother consensus structure.

From a broader perspective, this method belongs to the refinement-oriented branch of CA matrix-based ensemble clustering. The key idea of this branch is that pairwise co-occurrence is only a starting point. Link-based methods enrich the matrix by considering inter-cluster connection information [2], [3]. Path-based methods further exploit indirect relations, assuming that two samples may be related even when their direct co-occurrence is weak, as long as they are connected through reliable intermediate samples [4], [5]. Methods such as GoT combine path-based refinement with density-based growth, which is especially useful when the initial CA matrix is sparse or fragmented [6]. Compared with these methods, the self-enhancement framework is more explicitly optimization-driven because it separates matrix refinement, noise modeling, and high-confidence constraint preservation in one objective.

Another related direction formulates CA refinement as representation recovery. Robust low-rank learning assumes that a clean consensus structure should have a compact latent form and uses low-rank constraints to suppress noisy pairwise evidence [7]. Matrix-completion-based methods recover missing or unreliable similarities by exploiting the global structure of the matrix [8]. Semidefinite programming provides a global agreement optimization perspective [9], while weighted spectral or semi-supervised formulations introduce additional reliability or supervision into the consensus process [10], [11]. These methods differ in technical form, but they share the same motivation: the raw CA matrix should be corrected before it is used for final clustering.

Recent studies further show that pairwise refinement alone may still be insufficient. Since a CA matrix compresses an ensemble into pairwise averages, it may lose high-order relations among multiple samples, dissimilarity information, and global partition constraints. High-order learning and graph tensor learning attempt to recover structural information beyond independent pairwise entries [12], [13]. Latent space learning further maps consensus evidence into a hidden representation where clustering structure may be easier to separate [14]. SDGCA extends the standard CA view by jointly modeling similarity and dissimilarity, which is important because reliable non-cooccurrence can also provide useful clustering evidence [15]. TBAWEC introduces balance constraints into the consensus optimization, suggesting that the quality of an enhanced CA matrix should also be evaluated by the global properties of the resulting partition [16].

B. Supplementary Analysis of Graph-Based Ensemble Methods

Graph-based ensemble clustering aims to learn a consensus graph whose topology reflects the final partition structure. In this view, consensus learning is no longer treated as a separate post-processing step after graph construction. Instead, graph construction, graph refinement, and cluster-structure enforcement are jointly considered so that the learned graph can directly support partition extraction.

This idea is illustrated by the structured graph learning framework of Nie et al. [17]. Their model assigns a weight α_m to each base clustering and learns a weighted similarity graph by solving

$$\begin{aligned} \min_{\mathbf{S}, \alpha} \quad & \|\mathbf{S} - \mathbf{A}\|_F^2, \quad \mathbf{A} = \sum_{m=1}^M \alpha_m \mathbf{A}_m, \\ \text{s.t.} \quad & \mathbf{S}\mathbf{1} = \mathbf{1}, \mathbf{S} \geq 0, \alpha_m \geq 0, \alpha^\top \mathbf{1} = 1, \text{rank}(\mathbf{L}_\mathbf{S}) = n - K. \end{aligned}$$

Here, $\mathbf{A}_m = \mathbf{Y}_m \mathbf{Y}_m^\top$ is the similarity matrix induced by the m -th base clustering and $\mathbf{Y}_m$ is the corresponding cluster-indicator matrix. The matrix $\mathbf{L}_\mathbf{S}$ denotes the Laplacian matrix of $\mathbf{S}$. The key point is that the graph is constrained to be stochastic and nonnegative, while the rank condition enforces $\mathbf{S}$ to have exactly K connected components. Therefore, the learned graph is encouraged to carry the target clustering structure during optimization.

By relaxing the rank constraint through the Laplacian eigen-spectrum, the model can be rewritten as

$$\begin{aligned} \min_{\mathbf{S}, \mathbf{H}} \quad & \|\mathbf{S} - \mathbf{A}\|_F^2 + 2\gamma \text{Tr}(\mathbf{H}^\top \mathbf{L}_\mathbf{S} \mathbf{H}) \\ \text{s.t.} \quad & \mathbf{H}^\top \mathbf{H} = \mathbf{I}_K. \end{aligned}$$

Here, $\mathbf{H} \in \mathbb{R}^{n \times K}$ is the relaxed embedding matrix and $\mathbf{I}_K \in \mathbb{R}^{K \times K}$ is the identity matrix. This formulation unifies graph learning and graph partitioning in one model. The final clustering result can then be obtained from the connectivity structure of $\mathbf{S}$, rather than from an additional discretization procedure such as k -means.

Early graph-based ensemble clustering methods mainly relied on direct graph partitioning. Strehl and Ghosh [18] proposed several representative consensus functions based on similarity reclustering, hypergraph partitioning, and meta-cluster construction. Fern and Brodley [19] further introduced a hybrid bipartite graph formulation, where instances and clusters are jointly modeled as vertices in a bipartite graph. These methods established the basic graph perspective of ensemble clustering by converting base partitions into graph structures and then applying graph partitioning techniques.

Later work moved beyond direct partitioning by improving the reliability and expressiveness of the graph itself. Huang et al. [20] constructed a sparse reliable graph through elite neighbor selection and used random walks to derive probability-trajectory-based similarity. Zhou et al. [21] proposed an ensemble clustering method with adaptive multiplex graph refinement, where multiplex graphs and consensus representations are iteratively updated through diffusion. These studies show that graph quality is crucial for consensus learning, since unreliable or noisy edges may directly affect the final partition.

Recent graph-based methods further extend this idea by introducing richer graph structures. Some methods use tensorized or higher-order representations to capture dependencies that cannot be expressed by ordinary pairwise graphs [22]. Other methods focus on heterogeneous or multi-view settings by jointly integrating multiple graphs [23], [24]. Hypergraph-based methods have also attracted increasing attention because they can naturally model higher-order relations among samples and base clusters. Representative studies include structured hypergraph learning [25], dynamic anchor-based hypergraph reconstruction [26], and k -HyperEdge Medoids based consensus learning [27].

C. Supplementary Analysis of Probabilistic and Statistical Methods

Probabilistic and statistical methods model consensus formation through latent variables and uncertainty. A representative example is Bayesian Consensus Clustering by Lock and Dunson [28], which infers an overall clustering from multiple source-specific clusterings measured on the same N objects. For object n and source m , the model introduces an overall cluster indicator C_n and a source-specific indicator L_{mn} , where L_{mn} follows C_n only probabilistically. For each source m , observations follow a K -component mixture

$$X_{mn} \sim f_m(\cdot \mid \theta_{m, L_{mn}}), \quad L_{mn} \in \{1, \dots, K\}.$$

Dependence across sources is mediated by C_n through an adherence function

$$P(L_{mn} = k | C_n) = \gamma(k, C_n, \alpha_m),$$

where $\alpha_m \in [1/K, 1]$ controls how strongly source m follows the overall clustering [28]. Lock and Dunson [28] use the following form

$$\gamma(L_{mn}, C_n, \alpha_m) = \alpha_m \mathbb{I}\{L_{mn} = C_n\} + \frac{1 - \alpha_m}{K - 1} \mathbb{I}\{L_{mn} \neq C_n\},$$

so that $\alpha_m = P(L_{mn} = C_n)$. The overall mixing weights $\pi = (\pi_1, \dots, \pi_K)$ satisfy

$$\pi_k = P(C_n = k), \quad \pi \sim \text{Dirichlet}(\eta),$$

and α_m is given a Beta prior.

Posterior inference proceeds through Gibbs sampling. Two conditional updates show how the model combines evidence from different sources. The overall cluster for object n is updated using the current source-specific labels

$$P(C_n = k | \mathbf{L}_{\cdot n}, \pi, \alpha) \propto \pi_k \prod_{m=1}^M \gamma(L_{mn}, k, \alpha_m).$$

Each source-specific label is updated using the local likelihood and its adherence to C_n

$$P(L_{mn} = k | X_{mn}, C_n, \theta_{mk}, \alpha_m) \propto \gamma(k, C_n, \alpha_m) f_m(X_{mn} | \theta_{mk}).$$

This Bayesian formulation yields an explicit consensus clustering C , source-specific clusterings $L_1, \dots, L_M$, and learned adherence parameters α_m that quantify heterogeneity across sources [28]. More importantly, it captures the central idea of probabilistic ensemble clustering. The consensus partition is treated as a latent object, while uncertainty and source-specific variation are inferred jointly rather than handled through deterministic refinement rules. The broader literature can be understood from two related perspectives, namely model-based probabilistic inference and statistical analysis of stability and consistency.

1) Model-Based Probabilistic Consensus and Inference:

This line of work builds an explicit generative or latent-variable model for the ensemble itself. Rather than relying on simple voting, it assumes that the observed base clusterings are generated from an underlying latent partition and then uses statistical inference to recover that structure.

Early work by Topchy et al. [29], [30] introduced this perspective by formulating consensus clustering with mixture models. They transformed the problem into maximum-likelihood estimation for a finite mixture of multinomial distributions, which can be solved efficiently using the Expectation-Maximization algorithm. This formulation provides a unified way to handle partial clusterings and missing labels. Building on the same model-based view, Nguyen and Caruana [31] developed an Expectation-Maximization-like algorithm to find the median partition without computationally expensive pairwise similarity calculations. Wu et al. [32] later established a unified framework for k -means-based consensus clustering and showed that these methods essentially maximize a specific utility function. To improve robustness when a single fitted model may be inadequate, Casa et al. [33] introduced a density-based ensemble strategy with model averaging.

Bayesian modeling forms an important extension of this direction because prior distributions allow uncertainty and heterogeneity to be encoded explicitly. Lock and Dunson [28] jointly infer the consensus structure and source-specific parameters within a unified Bayesian framework. Lourenço et al. [34] introduced probabilistic consensus clustering and extended the classic evidence accumulation algorithm from a statistical perspective. More recently, probabilistic graphical models have also been used in ensemble clustering. Huang et al. [35] used factor graphs to address automatic cluster number determination and scalability, while Zhu et al. [36] combined probabilistic graphical modeling with graph regularization to reduce the sensitivity of single-cell data clustering to initialization.

2) *Statistical Stability, Resampling, and Theoretical Foundations:* A second line of research focuses less on building a full generative model and more on evaluating whether an ensemble result is stable, reliable, and theoretically well grounded. Statistical tools such as resampling, hypothesis testing, and formal analysis are used to examine the validity of cluster structure and to resolve label alignment and consistency issues within the ensemble process.

Resampling techniques are classic tools for assessing clustering stability. Levine and Domany [37] evaluated the validity of natural clusters by examining the consistency of results under data perturbation. Fischer and Buhmann [38] introduced bagging and bootstrap ideas into clustering and showed that data resampling can enhance the diversity of base clusterings and the robustness of the final result. Recent work by Nascimento et al. [39] extended bagging to three-dimensional statistical shape analysis, showing that this line of thinking remains useful for complex data.

The same perspective also motivates theoretical work on consistency and label correspondence. Jain [40] established the Mean Partition Theorem for consensus clustering, using orbit space theory to prove the existence of the Fréchet mean and thereby providing a foundation for median-based ensemble methods. To address the label alignment challenge, Ayad and Kamel [41], [42] recast relabeling as a multi-response regression problem. More recently, Alziati et al. [43] developed a synchronized relabeling framework that overcomes the suboptimality of traditional independent processing through coupled optimization. In parallel, He et al. [44] refined the theoretical treatment of categorical data clustering from an information-theoretic perspective through mutual information maximization.

D. Supplementary Analysis of Weighting-Based Ensemble Methods

Weighting-based ensemble methods are motivated by the fact that base clusterings are rarely equally informative. Some partitions are globally unreliable, some are redundant, and others preserve useful local structures even when their overall quality is limited. Therefore, these methods do not aggregate all ensemble members uniformly. Instead, they differentiate their contributions according to reliability, complementarity, or local confidence. Existing studies can be organized into two related directions. One direction removes noisy or repetitive members before consensus, while the other preserves richer ensemble evidence by assigning weights at different granularities.

1) *Ensemble Selection and Pruning:* Ensemble selection and pruning operate at the member level by retaining high-quality and complementary partitions. The basic idea is that removing weak or repetitive base clusterings may improve consensus quality more effectively than averaging all available members.

Early studies mainly designed criteria for identifying useful members. Zhou and Tang [45] showed that selective ensembles can outperform full ensembles by using mutual information to filter out low-quality partitions. Hong et al. [46] introduced a resampling-based selection framework that balances accuracy and diversity through a weighted fitness function. Later work refined the screening process with richer structural criteria. Akbari et al. [47] organized base partitions through hierarchical clustering to identify representative subsets. Yousefnezhad et al. [48] incorporated an independency measure and proposed an index that jointly evaluates quality, diversity, and independence. Khalili et al. [49] further developed a hierarchical selection algorithm based on an extended Jaccard measure.

More recent studies have moved from heuristic screening toward explicit optimization. Faceli et al. [50] used a multi-objective evolutionary algorithm to optimize consensus quality and the number of clusters simultaneously. Li et al. [51] applied evolutionary multi-objective pruning to single-cell RNA-seq analysis, showing that selection can be especially important in noisy biological settings. Shi et al. [52] introduced a transfer-learning-based selection method, where source-domain knowledge is used to guide ensemble selection in the target domain.

This optimization-oriented trend also encourages closer integration between selection and consensus. Valente de Oliveira et al. [53]

embedded pruning strategies into particle swarm clustering. Ma et al. [54] refined selection results by combining multiple algorithms and combination strategies. Banerjee et al. [55] further coupled ensemble selection and weighting within a unified iterative framework. On the practical side, Chiu and Talhouk [56] developed an R package that provides multiple selection and consensus strategies for applied clustering analysis.

2) *Multi-Granularity Weighting Strategies*: Weighting strategies take a complementary view. Instead of discarding members directly, they preserve more ensemble information and control how strongly different sources of evidence influence the final consensus. A clear pattern in the literature is the progressive refinement of weighting granularity, moving from partition-level weighting to cluster-level and object-level weighting.

Early weighting schemes mainly operated at the partition level, where each base clustering receives a single global reliability score. Al-Razgan and Domeniconi [57] explored this idea in subspace clustering through weighted algorithms and later formalized it as a weighted cluster ensemble framework using normalized mutual information to quantify partition reliability. Vega-Pons et al. [58] proposed a weighted partition consensus method based on kernels, combining partition validity indices with kernel-based consensus learning.

Once researchers recognized that reliable and unreliable clusters may coexist within the same partition, weighting naturally shifted to the cluster level. Rashidi et al. [59] designed an exponential dependability measure based on information theory and used it to construct a weighted CA matrix. Nazari et al. [60] provided a broader study of weighting strategies based on cluster quality and diversity. Wang et al. [61] adapted this idea to fuzzy clustering ensembles by estimating the unreliability of fuzzy clusters through entropy criteria. Xu and Ding [62] further proposed a dual-granularity method that jointly considers global cluster reliability and local sample similarity.

The finest level of refinement appears at the object level, where weighting is used to suppress local noise and sample-specific uncertainty. Ren et al. [63], [64] assigned weights to individual objects so that difficult or unstable instances exert less influence during consensus. Rashedi and Mirzaei [65] used a boosting-inspired mechanism to update sample weights dynamically in hierarchical cluster ensembles. Ji et al. [66] estimated sample certainty through binarized statistics and distinguished between core and halo samples in a multi-stage fusion process. Bai et al. [67] designed multiple weighted consensus metrics from an information-theoretic perspective to reduce uncertainty during aggregation.

E. Supplementary Analysis of Uncertainty-Based Ensemble Methods

Uncertainty-based ensemble methods focus on the ambiguity caused by noisy data, unstable samples, overlapping clusters, and inconsistent base partitions. Unlike hard consensus methods that force each object into a single cluster, these methods preserve uncertainty through soft memberships, boundary regions, belief assignments, or delayed decisions. The main idea is that ambiguity should not always be removed before consensus. In many cases, it provides useful information about sample reliability, cluster boundaries, and the confidence of ensemble evidence.

A representative example is the adaptive fuzzy framework of Yu et al. [68], which was designed for high-noise and high-dimensional cancer data. Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$ be the dataset, and suppose that M base clusterings are represented by soft membership matrices $\{\mathbf{U}^{(m)}\}_{m=1}^M$. This type of model jointly learns a consensus fuzzy partition $\mathbf{U}$, cluster prototypes $\mathbf{V} = \{\mathbf{v}_k\}_{k=1}^K$, and adaptive reliabilities w_m for base partitions. A typical objective can be written as

$$\begin{aligned} \min_{\mathbf{U}, \mathbf{V}, \mathbf{w}} \quad & \sum_{i=1}^N \sum_{k=1}^K u_{ik}^\gamma \|\mathbf{x}_i - \mathbf{v}_k\|_2^2 + \lambda \sum_{m=1}^M w_m \|\mathbf{U} - \mathbf{U}^{(m)}\|_F^2 \\ & + \eta \sum_{m=1}^M w_m \log w_m \\ \text{s.t.} \quad & u_{ik} \geq 0, \quad \sum_{k=1}^K u_{ik} = 1, \quad w_m \geq 0, \quad \sum_{m=1}^M w_m = 1. \end{aligned}$$

The first term preserves the fuzzy clustering structure of the data. The second term encourages the consensus membership matrix $\mathbf{U}$ to agree with the base memberships $\mathbf{U}^{(m)}$, while w_m controls the contribution of each base partition. The entropy-like regularizer prevents the weights from collapsing to a single partition. Under this formulation, the weight update usually takes a softmax-like form driven by the disagreement between the consensus and each base partition

$$w_m \propto \exp\left(-\frac{\lambda}{\eta} \|\mathbf{U} - \mathbf{U}^{(m)}\|_F^2\right), \quad \sum_{m=1}^M w_m = 1.$$

The prototype update follows the standard weighted fuzzy c -means form

$$\mathbf{v}_k = \frac{\sum_{i=1}^N u_{ik}^\gamma \mathbf{x}_i}{\sum_{i=1}^N u_{ik}^\gamma}.$$

These updates illustrate the basic mechanism of uncertainty-aware fuzzy consensus. Base partitions that agree better with the evolving consensus receive larger weights, while ambiguous samples remain softly assigned rather than being forced into hard labels.

Fuzzy-based soft consensus methods form the most direct way to represent uncertainty. They preserve membership degrees from base clusterings and therefore reduce the information loss caused by hard assignments. Sevillano et al. [69] introduced a soft consensus function based on the Borda count, where soft memberships are transformed into ranked preferences for fusion. Parvin and Minaei-Bidgoli [70] proposed a locally adaptive framework that selects and weights high-confidence fuzzy clusters to handle noise. Su et al. [71] further developed a hierarchical fuzzy ensemble that aggregates clusters by constructing a fuzzy similarity graph, which is suitable for large-scale data.

Several studies further extend fuzzy consensus to high-dimensional or structurally complex settings. Lahmar et al. [72], [73] combined spectral clustering with random projections to reduce sensitivity to initialization and dimensionality. Shi et al. [74] used spectral learning to align membership matrices in random-projection-based ensembles. Li et al. [75] introduced self-coassociation and prototype propagation to mine deeper structures from fuzzy ensemble evidence. Bian et al. [76] proposed a one-step fuzzy ensemble that embeds graphs with the ground-truth cluster number to handle inconsistent cluster counts across base members. Ren and Sun [77] designed a bi-directional fuzzy c -means ensemble that incorporates local information to improve the treatment of boundary samples.

Reliability assessment is another important issue in fuzzy ensembles. Bagherinia et al. [78] and Chen et al. [79] argued that fuzzy clusters have different levels of credibility and proposed reliability-based frameworks that weight consensus according to cluster quality measures such as compactness. Yeh and Yang [80] introduced a fuzzy generalized Rand index to evaluate fuzzy ensemble results. These studies show that fuzzy uncertainty is useful not only for representing ambiguous samples, but also for estimating the reliability of clusters and base partitions.

Beyond fuzzy logic, rough set theory and granular computing provide alternative tools for modeling boundary uncertainty. These methods describe uncertain objects through upper and lower approximations, which makes them suitable for data with unclear cluster boundaries. Saha et al. [81] proposed a rough-fuzzy framework for categorical data, where pure and semi-rough objects are distinguished to build a stable initial structure. Hu et al. [82], [83] formulated

ensemble clustering as a maximum granularity aggregation problem and developed incremental fuzzy-rough frameworks based on knowledge granulation. Xu and Ding [84] used uncertainty metrics from granular computing to screen base clusterings. Wu et al. [85] further refined this idea with a granularity-weighted ensemble that considers reliability at the partition, cluster, and sample levels.

Three-way decision theory provides another way to handle uncertain samples by allowing the algorithm to accept, reject, or defer decisions. This mechanism is useful when the available ensemble evidence is insufficient for a confident assignment. Wang and Chen [86] applied this idea to incomplete data with granular computing. Fan et al. [87], [88] developed density-sensitive and perturbation-based three-way ensemble methods to quantify sample instability. Jiang et al. [89] combined shadowed sets with three-way decisions to handle overlapping distributions, while Wu et al. [90] used random subspaces to express boundary uncertainty more effectively.

Evidence-theoretic methods provide a further perspective on uncertainty fusion by modeling incomplete and conflicting information. Masson and Denoeux [91] treated ensemble clustering as a belief assignment problem within a partition lattice. Li et al. [92] adopted a multi-granulation perspective based on evidence theory to overcome the limitations of simple voting. Fu et al. [93] proposed a stable clustering ensemble that models whether a sample belongs to a cluster, does not belong to it, or remains unknown.

F. Supplementary Analysis of Representation Enhancement Methods

Representation enhancement methods improve ensemble clustering by enriching the evidence used for consensus learning. Instead of only redesigning the aggregation rule, this line of work constructs more informative representations before or during consensus formation. These representations may take the form of latent features, graph embeddings, spectral structures, consensus-aware features, or neural representations. They are especially useful when raw features, base partitions, or simple pairwise summaries are insufficient to describe the underlying cluster structure.

One important branch uses deep models to generate latent representations for complex data. Geddes et al. [94] used deep autoencoders to compress sparse scRNA-seq data and then performed ensemble clustering in the latent space for cell-type identification. Xu et al. [95] employed stacked autoencoders for community detection, where nonlinear node representations are learned from multiple structural views. For sequential industrial data, Wang et al. [96] combined recurrent autoencoders with handcrafted features in a dual-stream diagnosis framework. These methods show that representation enhancement can improve the quality of the evidence supplied to consensus learning, especially for high-dimensional, sparse, sequential, or graph-structured data.

Later studies made representation generation more adaptive. Instead of learning a fixed latent space independently of clustering difficulty, these methods adjust the representation according to sample importance, reconstruction behavior, or ensemble diversity. Shi et al. [97] introduced a broad-learning autoencoder that re-weights samples during training so that difficult-to-cluster instances receive greater attention. Saronian et al. [98] encouraged diversity by combining multiple deep continuous clustering models with different reconstruction losses. In these methods, representation enhancement is no longer a simple preprocessing step, but a way to produce complementary clustering evidence.

Another branch enhances representations at the consensus level. These methods learn a space in which the relationships among samples and base clusterings can be modeled more flexibly than by voting, CA matrices, or hypergraph partitioning alone. Du and Wang [99] proposed a graph convolutional framework that encodes raw features and the ensemble CA matrix into a unified latent space through a variational graph autoencoder, thereby treating consensus construction as a generative representation learning problem. They further modeled coupling relationships among base clusterings with

an explicit generative formulation [100]. Gu et al. [101] also adopted graph convolutional embeddings and used supervised constraints from quantized industrial labels to refine the consensus graph. These methods connect sample-level features, ensemble-level evidence, and consensus structure within a unified representation.

Attention-based and broad-learning-based methods provide another form of representation enhancement. Their goal is to learn how different base clusterings, feature components, or cluster-level signals should contribute to the final consensus. Hao et al. [102] argued that CA matrices may lose high-order information and introduced an attentional network to estimate the importance of different base clusterings for each sample. Li and Shi [103] proposed a consensus broad learning network that maps feature-level and cluster-level information into a broad expansion space for efficient fusion. Compared with fixed fusion rules, these methods construct consensus-aware representations that better reflect heterogeneous contributions from the ensemble.

Recent work further weakens the separation between representation learning and consensus construction. Rather than generating features first and aggregating them afterward, these methods allow the consensus objective to guide representation learning directly. Liang et al. [104] proposed a joint deep ensemble framework that combines diverse feature extraction, dropout-based perturbation, and orthogonal rotation for cluster alignment in a differentiable architecture. Chen et al. [105] introduced contrastive ensemble clustering, where a contrastive regularizer aligns original features with global representations induced by the ensemble so that local data structure can be preserved during consensus learning. These studies suggest that representation enhancement is moving from independent feature construction toward joint optimization of representation and consensus.

G. Supplementary Analysis of Scalable Ensemble Methods

Scalable ensemble methods address the computational and memory bottlenecks that arise when ensemble clustering is applied to large-scale data. In many classical methods, the main cost comes from constructing and storing dense pairwise structures, especially the CA matrix, which usually requires $O(N^2)$ time and space. This quadratic dependence limits the use of ensemble clustering when the number of samples becomes large. Therefore, scalable methods aim to preserve useful consensus information while avoiding full pairwise computation. Existing studies mainly reduce the number of objects involved in consensus, approximate expensive optimization procedures, or distribute and accelerate the computation.

A common strategy is to replace original samples with representative units before consensus construction. Random-sample-partition-based methods reduce the scale of each base clustering problem by splitting the data into statistically representative subsets. Du et al. and Mahmud et al. [106], [107] adopted this idea to generate base clusterings from smaller partitions while maintaining the global distributional characteristics of the data. In this way, ensemble generation can be performed on reduced data blocks rather than the complete dataset, which lowers the cost of each clustering run.

Another representative reduction strategy is to use centroids, anchors, or micro-clusters as proxies for individual samples. Hore et al. [108] proposed a scalable framework that first clusters disjoint subsets and then represents each subset by centroids. The global consensus problem is then transformed into a smaller matching problem among representative centroids. Similarly, Liang et al. and Yu et al. [109], [110] used representative micro-clusters or cluster units to construct a compressed CA representation. Compared with the dense $N \times N$ CA matrix, this representation stores relations among a much smaller set of representative objects. These methods show that scalability can be achieved by shifting consensus learning from the sample level to the representative level.

For methods that still need to exploit global data structure, algorithmic approximation provides another solution. Spectral ensemble methods are often computationally expensive because they require eigendecomposition on large affinity or Laplacian matrices. He and Zhang [111] proposed iterative ensemble normalized cuts,

which approximate the global spectral solution by integrating results from small subsamples. Li et al. [112] developed a landmark-based spectral ensemble framework, where landmark selection is used in ensemble generation and a bipartite graph is used in consensus fusion. Zhang et al. [113] further introduced an anchor-based fast spectral ensemble method that performs singular value decomposition only once on a sparse anchor graph to generate diverse embeddings. These studies avoid repeated spectral decomposition on dense matrices and therefore make spectral ensemble clustering more practical for large datasets.

Other approximation methods accelerate ensemble clustering by simplifying the feature space or the manifold structure. Rathore et al. [114] combined random projection with visual assessment techniques to reduce the cost of processing high-dimensional data. Xu et al. [115] integrated k -means with Laplacian Eigenmaps to approximate manifold information efficiently for sparse document clustering. Although these methods differ in implementation, they share the same principle. Expensive global structures are replaced by lower-dimensional, sparse, or approximate representations that retain the information needed for consensus learning.

When the data size exceeds the capacity of a single machine, scalability also depends on distributed computation and hardware-level acceleration. Li et al. [116] implemented a parallel co-clustering ensemble system based on a distributed multi-objective genetic model, where sub-populations evolve in parallel across different nodes to optimize the consensus objective. Ren and Sun [77] proposed a decentralized framework in which agents exchange cluster prototypes instead of raw data, reducing communication cost and improving privacy preservation. Porcello [117] explored hardware acceleration by implementing ensemble clustering on field-programmable gate arrays, where voting and grid-based clustering operations can be parallelized at the circuit level. These methods extend scalable ensemble clustering from algorithmic approximation to system-level execution.

II. BENCHMARK DATASETS

This section summarizes the benchmark datasets used in the experimental study. Table S1 reports the identifier, number of instances, features, and classes for each dataset, followed by brief descriptions of their sources and data characteristics.

We briefly introduce the 86 datasets used in our experiments below.

- **20NG**¹ is a high-dimensional text benchmark derived from the 20 Newsgroups corpus, containing documents from 20 discussion categories and commonly used in text clustering and document classification studies.
- **Abalone**² contains physical measurements of abalone specimens, such as length, diameter, height, and shell-related attributes, and is widely used in clustering, regression, and life-stage analysis.
- **Absenteeism at work**³ records employee absenteeism together with personal, work-related, and health-related variables, and is useful for behavior analysis and occupational data mining.
- **Aggregation**⁴ is a classical two-dimensional synthetic clustering benchmark composed of several non-convex groups, and is often used to evaluate the ability of clustering methods to capture irregular cluster shapes.
- **AIDS Clinical Trial Group Study 175**⁵ is a clinical dataset related to HIV/AIDS studies, containing patient-level medical indicators, and is commonly used in medical prediction and biomedical data analysis.

- **ALLAML**⁶ is a high-dimensional gene expression dataset designed to distinguish acute lymphoblastic leukemia from acute myeloid leukemia, and is frequently used in feature selection and bioinformatics studies.
- **Appendicitis**⁷ is a small medical diagnosis dataset containing clinical attributes for appendicitis-related decision making, and is often used as a compact benchmark in classification and clustering experiments.
- **Arcene**⁸ is an artificial high-dimensional benchmark originally proposed for feature selection, containing informative and distracting variables for binary pattern recognition.
- **Auction Verification**⁹ is a verification-oriented dataset related to auction or transaction analysis, and is commonly used in binary prediction and anomaly-related studies.
- **Balance**¹⁰ contains balance-scale configurations described by physical measurements, and categorizes samples into left, right, and balanced classes.
- **BASEHOCK**¹¹ is a text benchmark derived from the 20 Newsgroups corpus, distinguishing baseball and hockey topics, and is often used in sparse high-dimensional text analysis.
- **BBC**¹² is a news-document text benchmark composed of BBC articles from five topical categories, and is widely used in document clustering and topic discovery studies.
- **Binalpha**¹³ is derived from the Binary Alphadigits dataset, containing binary images of digits and uppercase letters, and is often used in image clustering and character recognition benchmarks.
- **Breast (Diagnostic)**¹⁴ contains diagnostic breast cancer measurements computed from digitized images of fine needle aspirates, and is widely used for benign/malignant discrimination.
- **Breast Cancer Wisconsin**¹⁵ is a classic medical benchmark for breast cancer diagnosis, containing cell-level attributes and binary class labels.
- **Caltech20**¹⁶ is a 20-class subset of Caltech101, consisting of object images from multiple semantic categories, and is commonly used in image clustering and multi-view learning.
- **Cardiotocography_10classes**¹⁷ is the 10-class version of the cardiotocography dataset, reflecting detailed fetal state categories derived from clinical monitoring signals.
- **Cardiotocography_3classes**¹⁸ is the 3-class version of the cardiotocography dataset, grouping fetal states into broader categories for standard classification and clustering benchmarks.
- **Chess**¹⁹ is a symbolic benchmark derived from chess configurations or endgame states, and is widely used in rule discovery and structured pattern learning.
- **CLL_SUB_111**²⁰ is a chronic lymphocytic leukemia gene expression subset, and is frequently adopted in high-dimensional biomedical feature selection research.
- **CNAE-9**²¹ contains Brazilian business text documents grouped into nine economic activity categories, and is a representative

⁶<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/ALLAML.mat>

⁷<https://sci2s.ugr.es/keel/dataset/data/classification/appendicitis.zip>

⁸<https://archive.ics.uci.edu/dataset/167/arcene>

⁹<https://archive.ics.uci.edu/dataset/713/auction+verification>

¹⁰<https://archive.ics.uci.edu/dataset/12/balance+scale>

¹¹<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/BASEHOCK.mat>

¹²<http://mlg.ucd.ie/datasets/bbc.html>

¹³https://www.tensorflow.org/datasets/catalog/binary_alpha_digits

¹⁴<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

¹⁵<https://archive.ics.uci.edu/dataset/15/breast+cancer+wisconsin+original>

¹⁶<https://data.caltech.edu/records/mzrqj-6wc02>

¹⁷<https://archive.ics.uci.edu/dataset/193/cardiotocography>

¹⁸<https://archive.ics.uci.edu/dataset/193/cardiotocography>

¹⁹<https://sci2s.ugr.es/keel/dataset/data/classification/chess.zip>

²⁰<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/CLL-SUB-111.mat>

²¹<https://archive.ics.uci.edu/dataset/233/cnae+9>

¹<http://qwone.com/~jason/20Newsgroups/>

²<https://archive.ics.uci.edu/dataset/1/abalone>

³<https://archive.ics.uci.edu/dataset/445/absenteeism+at+work>

⁴<http://cs.uef.fi/sipu/datasets/>

⁵<https://archive.ics.uci.edu/dataset/890/aids+clinical+trials+group+study+175>

TABLE S1
DETAILS OF DIFFERENT DATASETS

ID	Dataset	Instance	Feature	Class	ID	Dataset	Instance	Feature	Class
D1	20NG	18846	26214	20	D44	Optdigits	5620	64	10
D2	Abalone	4177	8	3	D45	ORL	400	1024	40
D3	Absenteeism at work	740	19	19	D46	orlraws10P	100	10304	10
D4	Aggregation	788	2	7	D47	Page Blocks	5473	10	5
D5	AIDS Clinical Trial Group Study 175	2139	23	2	D48	PCMAC	1943	3289	2
D6	ALLAML	72	7129	2	D49	PD	10992	16	10
D7	Appendicitis	106	7	2	D50	Phishing	2456	30	2
D8	Arcene	200	10000	2	D51	Prostate_GE	102	5966	2
D9	Auction Verification	2043	7	2	D52	RELATHE	1427	4322	2
D10	Balance	625	4	3	D53	Rice	3810	7	2
D11	BASEHOCK	1993	4862	2	D54	Ring	1000	2	2
D12	BBC	2225	9635	5	D55	Room Occupancy Estimation	10129	16	4
D13	Binalpha	1404	320	36	D56	Seed	210	7	3
D14	Breast (Diagnostic)	569	30	2	D57	Seismic Bumps	2584	18	2
D15	Breast Cancer Wisconsin	699	9	2	D58	Semeion	1593	256	10
D16	Caltech20	2386	30000	20	D59	Sensor_Readings_2	5456	2	4
D17	Cardiotocography_10classes	2126	21	10	D60	Sensor_Readings_24	5456	24	4
D18	Cardiotocography_3classes	2126	21	3	D61	Sensor_Readings_4	5456	4	4
D19	Chess	3196	36	2	D62	SMK_CAN_187	187	19993	2
D20	CLL_SUB_111	111	11340	3	D63	SPF	1941	27	7
D21	CNAE-9	1080	856	9	D64	Statlog (Landsat Satellite)	6435	36	7
D22	COIL20	1440	1024	20	D65	Students' Dropout and Academic Success	4424	36	3
D23	Crowdsourced Mapping	10546	29	6	D66	Texture	5500	40	11
D24	Ecoli	336	7	8	D67	Thyroid	7200	5	3
D25	FCT	3780	54	7	D68	TOX_171	171	5748	4
D26	Fertility	100	9	2	D69	TUANDROMD	4464	241	2
D27	Foresttype	523	27	4	D70	UMIST	575	644	20
D28	Gender Gap in Spanish WP	4746	18	3	D71	USPS	11000	256	10
D29	GLIOMA	50	4434	4	D72	Vehicle	846	18	4
D30	Heart Failure Clinical Records	299	12	2	D73	warpAR10P	130	2400	10
D31	HTRU2	17898	8	2	D74	warpPIE10P	210	2420	10
D32	Image Segmentation	2310	19	7	D75	Waveform	5000	21	3
D33	Iranian Churn	3150	13	2	D76	Waveform Noise (Version1)	5000	40	3
D34	ISOLET	7797	617	26	D77	Wdbc	569	30	2
D35	Land Mines	338	3	5	D78	Website Phishing	1353	9	3
D36	Letter Recognition	20000	16	26	D79	WFRN	5456	24	4
D37	Lung	203	3312	5	D80	Wholesale Customers	440	7	3
D38	Madelon	2600	500	2	D81	Wine	178	13	3
D39	Maternal Health Risk	1014	6	3	D82	Wine Quality	6497	12	2
D40	MEU-Mobile KSD	2856	71	56	D83	Wireless Indoor Localization	2000	7	4
D41	MF	2000	649	10	D84	Yale	165	1024	15
D42	MNIST	5000	784	10	D85	Yeast	1484	8	10
D43	Obesity	2111	16	7	D86	Zoo	101	16	7

sparse text benchmark.

- **COIL20**²² consists of images from 20 object categories captured under different viewing angles, and is widely used in image clustering, subspace learning, and view-invariant recognition.
- **Crowdsourced Mapping**²³ is a mapping-related dataset constructed from crowdsourced information, and is useful for geographic classification and spatial data analysis.
- **Ecoli**²⁴ is a biological dataset for protein localization in *Escherichia coli*, where samples are grouped into multiple cellular destination classes.
- **FCT**²⁵ is a subset of the UCI Forest CoverType dataset, which contains cartographic variables for predicting forest cover types, and is commonly used in clustering, classification, and large-scale learning studies.
- **Fertility**²⁶ contains demographic, lifestyle, and medical attributes related to semen quality diagnosis, and is used as a compact benchmark for fertility-related pattern analysis.
- **Foresttype**²⁷ is a forest type mapping dataset from remote sensing studies, and is used for ecological recognition and land-cover analysis.

- **Gender Gap in Spanish WP**²⁸ contains statistics related to gender representation in Spanish Wikipedia articles, and is useful for bias analysis and social data mining.
- **GLIOMA**²⁹ is a glioma-related gene expression dataset for tumor subtype analysis, and is a typical high-dimensional biomedical benchmark.
- **Heart Failure Clinical Records**³⁰ contains clinical measurements associated with heart failure prognosis, and is widely used in medical risk prediction.
- **HTRU2**³¹ is a pulsar candidate dataset extracted from radio signal observations, and is widely used in binary classification and imbalance learning.
- **Image Segmentation**³² is a region-level image dataset, where each sample corresponds to an image segment described by low-level visual features.
- **Iranian Churn**³³ is a customer churn benchmark containing user-related behavioral and transactional attributes for attrition analysis.
- **ISOLET**³⁴ is an isolated spoken letter recognition dataset containing 26 alphabet classes, and is a standard benchmark for speech-related learning tasks.

²²<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/COIL20.mat>

²³<https://archive.ics.uci.edu/dataset/400/crowdsourced+mapping>

²⁴<https://archive.ics.uci.edu/dataset/39/ecoli>

²⁵<https://archive.ics.uci.edu/ml/datasets/covertypes>

²⁶<https://archive.ics.uci.edu/dataset/244/fertility>

²⁷<https://archive.ics.uci.edu/dataset/333/forest+type+mapping>

²⁸<https://archive.ics.uci.edu/dataset/852/gender%2Bgap%2Bin%2Bspanish%2Bwp>

²⁹<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/GLIOMA.mat>

³⁰<https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records>

³¹<https://archive.ics.uci.edu/dataset/372/htru2>

³²<https://archive.ics.uci.edu/dataset/50/image+segmentation>

³³<https://archive.ics.uci.edu/dataset/563/iranian+churn+dataset>

³⁴<https://archive.ics.uci.edu/dataset/54/isolet>

- **Land Mines**³⁵ is a mine-detection related dataset used for binary recognition tasks under noisy or uncertain conditions.
- **Letter Recognition**³⁶ refers to the Letter Recognition dataset, which contains samples of handwritten capital letters from 26 classes.
- **Lung**³⁷ is a high-dimensional lung cancer gene expression dataset, and is often adopted in biomedical feature selection studies.
- **Madelon**³⁸ is an artificial high-dimensional benchmark containing informative, redundant, and noisy variables, and is frequently used to test feature selection robustness.
- **Maternal Health Risk**³⁹ contains clinical indicators related to maternal health, and is used for risk assessment and healthcare analysis.
- **MEU-Mobile KSD**⁴⁰ is a benchmark related to mobile or knowledge-state diagnosis analysis, and is useful in structured prediction studies.
- **MF**⁴¹ refers to the Multiple Features handwritten digit dataset, which provides several feature views for the same digit instances.
- **MNIST**⁴² is a standard handwritten digit benchmark containing grayscale images of digits from 0 to 9, and is widely used in representation learning and clustering research.
- **Obesity**⁴³ is a health-related dataset for estimating obesity levels from eating habits and physical condition variables.
- **Optdigits**⁴⁴ is an optical handwritten digit recognition dataset with 10 digit classes and is widely used in image-based pattern recognition.
- **ORL**⁴⁵ is a face image dataset with variations in expression, illumination, and appearance, and is a classical benchmark in face clustering.
- **orlraws10P**⁴⁶ is a raw-image subset derived from ORL, often used in feature selection and subspace clustering studies.
- **Page Blocks**⁴⁷ is a document layout dataset for page block classification, where samples describe blocks extracted from page images.
- **PCMAC**⁴⁸ is a text benchmark derived from the 20 Newsgroups corpus, distinguishing PC and Mac related discussions.
- **PD**⁴⁹ is a shorthand benchmark corresponding to pen-based handwritten digit recognition data.
- **Phishing**⁵⁰ is a cybersecurity dataset used to distinguish phishing websites from legitimate ones based on webpage attributes.
- **Prostate_GE**⁵¹ is a prostate cancer gene expression dataset, and is widely used in high-dimensional biomedical analysis.

- **RELATHE**⁵² is a text benchmark derived from the 20 Newsgroups corpus, distinguishing religion and atheism related topics.
- **Rice**⁵³ contains rice grain measurements for variety recognition, and is frequently used in agricultural pattern analysis.
- **Ring**⁵⁴ is a synthetic benchmark with ring-like structure, and is commonly used to evaluate the ability of algorithms to separate non-linearly distributed clusters.
- **Room Occupancy Estimation**⁵⁵ contains environmental sensor data such as temperature, humidity, light, and CO₂, and is used for occupancy analysis.
- **Seed**⁵⁶ is a wheat seed dataset organized into three varieties, each described by geometric and shape-related measurements.
- **Seismic Bumps**⁵⁷ is a mining-related dataset used for assessing seismic hazard and dangerous underground activity conditions.
- **Semeion**⁵⁸ contains handwritten digit images represented by binary pixel features and is often used in digit classification benchmarks.
- **Sensor_Readings_2**⁵⁹ is a two-feature version of a wall-following robot navigation benchmark used in sensor-based classification.
- **Sensor_Readings_24**⁶⁰ is the full 24-feature version of the wall-following robot navigation dataset.
- **Sensor_Readings_4**⁶¹ is a reduced four-feature version of the wall-following robot navigation benchmark.
- **SMK_CAN_187**⁶² is a cancer-related high-dimensional gene expression benchmark used in feature selection and clustering studies.
- **SPF**⁶³ is a benchmark dataset frequently used in feature evaluation and clustering experiments.
- **Statlog (Landsat Satellite)**⁶⁴ is a remote sensing dataset with multiple land-cover classes described by spectral values.
- **Students' Dropout and Academic Success**⁶⁵ contains academic, demographic, and socioeconomic information and is used for analyzing student educational outcomes.
- **Texture**⁶⁶ is a visual benchmark used for texture recognition and image pattern analysis, with multiple texture-related classes.
- **Thyroid**⁶⁷ is a medical benchmark for thyroid disorder analysis and is widely used in healthcare-related learning tasks.
- **TOX_171**⁶⁸ is a toxicology-related high-dimensional gene expression dataset for biomedical learning and feature analysis.
- **TUANDROMD**⁶⁹ is an Android malware benchmark dataset used in cybersecurity classification and clustering studies.

³⁵<https://archive.ics.uci.edu/dataset/763/land+mines-1>

³⁶<https://archive.ics.uci.edu/dataset/59/letter+recognition>

³⁷<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/lung.mat>

³⁸<https://archive.ics.uci.edu/dataset/171/madelon>

³⁹<https://archive.ics.uci.edu/dataset/863/maternal+health+risk>

⁴⁰<https://archive.ics.uci.edu/dataset/399/meu+mobile+ksd>

⁴¹<https://archive.ics.uci.edu/dataset/72/multiple+features>

⁴²<https://www.dropbox.com/scl/fi/eiszho6pkht90pzb3xq5/CodeData15.zip?rlkey=dgithdekogojntwuyep1p0q8a&e=1&dl=0>

⁴³<https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition>

⁴⁴<https://archive.ics.uci.edu/dataset/80/optdigits>

⁴⁵<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/ORL.mat>

⁴⁶<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/orlraws10P.mat>

⁴⁷<https://archive.ics.uci.edu/dataset/78/page+blocks+classification>

⁴⁸<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/PCMAC.mat>

⁴⁹<https://archive.ics.uci.edu/dataset/81/pen+based+recognition+of+handwritten+digits>

⁵⁰<https://archive.ics.uci.edu/dataset/327/phishing+websites>

⁵¹<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/Prostate-GE.mat>

⁵²<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/RELATHE.mat>

⁵³<https://archive.ics.uci.edu/dataset/545/rice+cammeo+and+osmancik>

⁵⁴<https://sci2s.ugr.es/keel/dataset/data/classification/ring.zip>

⁵⁵<https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation>

⁵⁶<https://archive.ics.uci.edu/dataset/236/seeds>

⁵⁷<https://archive.ics.uci.edu/dataset/266/seismic+bumps>

⁵⁸<https://archive.ics.uci.edu/dataset/178/semeion+handwritten+digit>

⁵⁹<https://archive.ics.uci.edu/dataset/194/wall+following+robot+navigation+data>

⁶⁰<https://archive.ics.uci.edu/dataset/194/wall+following+robot+navigation+data>

⁶¹<https://archive.ics.uci.edu/dataset/194/wall+following+robot+navigation+data>

⁶²<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/SMK-CAN-187.mat>

⁶³<https://archive.ics.uci.edu/dataset/198/steel+plates+faults>

⁶⁴<https://archive.ics.uci.edu/dataset/146/statlog+landsat+satellite>

⁶⁵<https://archive.ics.uci.edu/dataset/697>

⁶⁶<https://sci2s.ugr.es/keel/dataset/data/classification/texture.zip>

⁶⁷<https://archive.ics.uci.edu/dataset/102/thyroid+disease>

⁶⁸<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/TOX-171.mat>

⁶⁹[https://archive.ics.uci.edu/dataset/855/tuandromd+\(tezpur+university+android+malware+dataset\)](https://archive.ics.uci.edu/dataset/855/tuandromd+(tezpur+university+android+malware+dataset))

- **UMIST**⁷⁰ is a face image dataset containing multiple individuals under varying poses and viewpoints.
- **USPS**⁷¹ is a handwritten digit benchmark widely used in recognition, representation learning, and clustering studies.
- **Vehicle**⁷² contains vehicle silhouette features and is used to recognize different vehicle types from shape-related attributes.
- **warpAR10P**⁷³ is a face-image benchmark frequently used in feature selection and clustering studies.
- **warpPIE10P**⁷⁴ is a benchmark derived from PIE face images and is often adopted in clustering and subspace learning studies.
- **Waveform**⁷⁵ is a synthetic waveform benchmark with three classes, designed for testing pattern recognition methods.
- **Waveform Noise (Version1)**⁷⁶ is the noisy version of the waveform benchmark and is commonly used to evaluate robustness under irrelevant features.
- **Wdbc**⁷⁷ is the shorthand form of the Wisconsin Diagnostic Breast Cancer dataset.
- **Website Phishing**⁷⁸ is a website security benchmark related to phishing or malicious webpage detection based on structural or content features.
- **WFRN**⁷⁹ is a robot navigation benchmark associated with wall-following sensor readings.
- **Wholesale Customers**⁸⁰ contains annual spending records of wholesale clients across several product categories.
- **Wine**⁸¹ contains chemical analysis results of wines derived from three cultivars grown in the same region of Italy.
- **Wine Quality**⁸² contains physicochemical measurements of red and white wines for quality evaluation and preference analysis.
- **Wireless Indoor Localization**⁸³ contains signal-strength measurements collected at different indoor positions for localization and classification tasks.
- **Yale**⁸⁴ is a widely used face dataset containing multiple subjects under varying expressions and illumination conditions.
- **Yeast**⁸⁵ is a biological dataset for protein localization prediction in yeast cells, containing multiple subcellular location categories.
- **Zoo**⁸⁶ contains animal descriptions with Boolean and categorical attributes, and is divided into seven animal-type classes.

III. METHOD PARAMETER SETTINGS

This section reports the hyperparameter settings used in the benchmark and complements the accuracy and runtime results in the main paper. When the original papers recommended fixed values, we adopted them directly. When candidate sets or search intervals

were provided, we tuned the corresponding parameters within the reported ranges, with modest narrowing only when exhaustive search was computationally prohibitive. In Table S2, “None” indicates that no additional manually tuned hyperparameter was introduced beyond the standard execution setting. The notations $[a, b]$ and a, b, c denote continuous ranges and discrete candidates evenly spaced values with step size s , respectively. These conventions clarify both the adopted parameter choices and the relative calibration burden of different method families.

IV. RUNTIME ANALYSIS

This section reports the computational cost of the 76 evaluated method variants on all 86 datasets. Table S3 summarizes the average, median, and maximum running times of each method, together with the number of time-out cases under the 3600-second limit. To provide a more fine-grained view, Tables S4–S7 further report the per-dataset running times in four landscape tables, covering datasets D1–D21, D22–D42, D43–D64, and D65–D86, respectively. In these tables, each row corresponds to a method ID, each column corresponds to a dataset ID, and each cell reports the average running time in seconds under the unified benchmark protocol. The notation “TO” indicates that the run reached the 3600-second limit. Dataset identifiers follow the order in Table S1, and method identifiers are consistent with Table S3.

The summary statistics characterize not only the overall computational cost of each method, but also the stability of that cost across datasets with different scales, feature dimensions, and clustering structures. The results reveal a clear efficiency hierarchy. Representation enhancement methods and scalable ensemble methods are generally the fastest and most stable groups, while many classical graph-based methods also remain computationally efficient. In contrast, CA-matrix-based and weighting-based methods exhibit larger runtime variation. These categories include both lightweight variants and optimization-intensive models whose costs can increase to hundreds or even thousands of seconds.

The gap between average and median running times provides useful information about runtime robustness. A large gap indicates that a method is efficient on most datasets but may slow down substantially on a small number of difficult cases, whereas a high median suggests consistently heavy computation. Time-out cases are concentrated in a limited subset of methods, reflecting the practical trade-off between expressive consensus modeling and scalability.

Compared with the summary statistics, the complete runtime matrix provides a more detailed view of robustness and scalability. Row-wise reading shows whether a method degrades smoothly across datasets or suffers abrupt scalability breakdowns. Column-wise reading shows whether a slowdown is method-specific or shared by several methods on the same dataset. This distinction helps diagnose potential computational bottlenecks, such as dense matrix construction or iterative optimization, and supports method selection under different runtime budgets. The matrix also helps determine whether an unfavorable average running time is caused by broadly distributed cost or by a small number of pathological method–dataset combinations. It therefore provides useful context for identifying outliers before drawing conclusions about the computational behavior of a method family.

⁷⁰<https://opendatalab.org.cn/OpenDataLab/UMIST>

⁷¹<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/USPS.mat>

⁷²<https://sci2s.ugr.es/keel/dataset/data/classification/vehicle.zip>

⁷³<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/warpAR10P.mat>

⁷⁴<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/warpPIE10P.mat>

⁷⁵<https://archive.ics.uci.edu/dataset/107/waveform+database+generator+version+1>

⁷⁶<https://archive.ics.uci.edu/dataset/107/waveform+database+generator+version+1>

⁷⁷<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

⁷⁸<https://archive.ics.uci.edu/dataset/379/website+phishing>

⁷⁹<https://archive.ics.uci.edu/dataset/194/wall+following+robot+navigation+data>

⁸⁰<https://archive.ics.uci.edu/dataset/292/wholesale+customers>

⁸¹<https://archive.ics.uci.edu/dataset/109/wine>

⁸²<https://archive.ics.uci.edu/dataset/186/wine+quality>

⁸³<https://archive.ics.uci.edu/dataset/422/wireless+indoor+localization>

⁸⁴<https://github.com/jundongli/scikit-feature/blob/master/skfeature/data/Yale.mat>

⁸⁵<https://archive.ics.uci.edu/dataset/110/yeast>

⁸⁶<https://archive.ics.uci.edu/dataset/111/zoo>

TABLE S2
PARAMETER SETTINGS OF DIFFERENT ENSEMBLE CLUSTERING METHODS

Category	Method	Venue	Parameter Setup
CA Matrix-based methods	[118]	Information Fusion	None
	[13]	AAAI	$\lambda \in [0.01, 0.2]$; $\gamma \in [0.1, 50]$; NNRate = 0.5; $o = 2$
	[119]	AAAI	$\lambda = 0.002$
	[120]	IEEE TPAMI	None
	[1]	IEEE TNNLS	$\alpha \in [0.7, 0.8]$; $\lambda \in [10^{-3}, 10^2]$
	[121]	Neurocomputing	$\lambda \in [10^{-3}, 10^3]$ on a log scale
	[12]	ACM MM	$\lambda_1 = 0.05$; $\lambda_2 = 7$
	[6]	AAAI	None
	[2]	Bioinformatics	DC = 0.9
	[7]	ACM CIKM	$\lambda_1, \lambda_2 \in [10^{-4}, 10^0]$ on a log scale
	[15]	IEEE TKDE	$\lambda = [0.06, 0.09]$; $\eta = [0.6, 1]$; $\theta = [0.6, 1]$
	[10]	IEEE TKDE	None
	[16]	IEEE TCSVT	$\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; $\gamma \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$
	[4]	Pattern Recognition	None
	[122]	LNCS	None
	[123]	IJCAI	$\lambda_1 = 1$; $\lambda_2 = 0.01$
Graph-based ensemble methods	[124]	ACM TKDD	$\lambda = [0.1 : 0.1 : 0.5]$
	[124]	ACM TKDD	$\lambda = [0.01 : 0.03 : 0.1]$
	[125]	JIIS	None
	[21]	IEEE TKDE	$\alpha = 0.85$; $k \in \{10, 20\}$
	[25]	Information Fusion	$\lambda \in [10^{-3}, 10^3]$ on a log scale
	[126]	Neurocomputing	$\beta = 2$; $\alpha = 0.5$
	[127]	IEEE TSMCS	Random walk step length $t = 20$
	[20]	IEEE TKDE	$K = \lfloor \sqrt{n}/2 \rfloor$; $T = \lfloor \sqrt{n}/2 \rfloor$
Probabilistic and statistical methods	[128]	ACM TKDD	$\gamma_1 \in (10^{-6}, 10^0)$ and $\gamma_2 \in (10^{-3}, 10^3)$ on a log scale
	[19]	ICML	None
	[23]	AAAI	$\gamma \in [10^{-5}, 10^5]$ on a log scale
	[26]	IJCAI	The number of anchors = $[k - 1, k : k + 3]$; $\alpha = [0.9, 1]$
	[29]	ICDM	None
	[30]	IEEE TPAMI	None
Weighting-based ensemble methods	[41]	IEEE TPAMI	None
	[35]	Pattern Recognition	$\tau = \sqrt{N}/2$
	[129]	IEEE TPAMI	$\lambda_1 = 0.001$; $\lambda_2 \in [10^{-5}, 10^1]$ on a log scale
	[130]	ICML	$\alpha = [0.05 : 0.05 : 1]$; $\lambda = 1$
	[32]	IEEE TKDE	None
Uncertainty-based methods	[131]	Pattern Recognition Letters	Bagging sampling ratio = 0.5
	[132]	IEEE TNNLS	$\lambda = 0.5$; $NB = 10$
	[133]	IEEE TCYB	$\theta = [0.2 : 0.1 : 1]$
	[134]	IEEE TCYB	upKNN = 20; lowKNN = 5; $\tau = 0.5$; $\mu = [0.2, 0.8]$
	[135]	Information Fusion	None
	[136]	IEEE TNNLS	$\gamma \in \{0, 0.01M^2, \dots, 0.81M^2\}$
	[58]	Pattern Recognition	$T_0 = 10^2$; $b = 0.98$; $r_{\max} = 10^4$
	[63]	ICDM	None
Representation enhancement methods	[137]	Bioinformatics	None
	[138]	Applied Artificial Intelligence	None
	[139]	Artificial Intelligence in Medicine	$\varepsilon = 0.2$; $m = 1.1$
	[82]	KBS	None
	[92]	Information Sciences	None
Scalable ensemble methods	[78]	FSS	None
	[75]	IEEE TFS	None
	[105]	IEEE TNNLS	$\alpha = 10^{-5}, \dots, 10^0$, $k = 5, \dots, 30$
	[140]	PR	$k = \{1, \dots, 10\}$
	[141]	Bioinformatics	$m = \{1000, \dots, 5000\}$
	[104]	ICME	None
	[142]	Information Sciences	None
	[98]	IEEE TITS	$k = 1, \dots, 10$
	[108]	Pattern Recognition	None
	[110]	IJAR	$(\alpha, \beta) \in [(0.5, 0.1), (0.9, 0.5)]$
	[143]	Artificial Intelligence Review	$\alpha = 0.3$
	[113]	Information Fusion	Anchors = $[6 : \min(12, \lfloor \log_2(n) \rfloor)]$; the nearest anchor = 5
	[144]	IEEE TKDE	None
	[145]	IEEE TKDE	None

TABLE S3

RUNNING TIME STATISTICS OF 76 EVALUATED METHOD VARIANTS ON 86 DATASETS. AVG., MEDIAN, AND MAX. DENOTE THE AVERAGE, MEDIAN, AND MAXIMUM RUNNING TIME, RESPECTIVELY. #TO DENOTES THE NUMBER OF TIME-OUT CASES (TIME $\geq$ 3600s).

ID	Category	Method	Venue	Avg.	Median	Max.	#TO
M1	CA Matrix-based methods	[118]	Information Fusion	0.77	0.65	5.58	0
M2		[13]	AAAI	365.24	27.32	1807.36	0
M3		[13]	AAAI	368.12	27.26	1846.77	0
M4		[119]	AAAI	747.86	81.45	1869.73	0
M5		[119]	AAAI	750.87	80.65	1880.71	0
M6		[120]	IEEE TPAMI	1.50	0.46	20.58	0
M7		[1]	IEEE TNNLS	67.91	13.04	1404.27	0
M8		[121]	Neurocomputing	12.52	2.85	276.32	0
M9		[12]	ACM MM	1207.38	422.88	TO	20
M10		[6]	AAAI	3192.45	3600.00	TO	75
M11		[2]	Bioinformatics	21.50	5.01	306.43	0
M14		[7]	ACM CIKM	79.63	10.07	1830.05	0
M15		[7]	ACM CIKM	76.83	9.63	1769.95	0
M16		[15]	IEEE TKDE	259.28	41.36	TO	2
M17		[10]	IEEE TKDE	0.23	0.21	1.02	0
M18		[16]	IEEE TCSVT	368.15	32.70	TO	4
M19		[4]	Pattern Recognition	1489.48	75.17	TO	35
M20		[122]	LNCS	10.97	4.69	147.74	0
M21		[123]	IJCAI	1497.12	138.01	TO	35
M22		[124]	ACM TKDD	548.28	106.19	TO	5
M23		[124]	ACM TKDD	1539.87	315.30	TO	35
M24		[125]	JIIS	2.57	0.65	47.64	0
M25	Graph-based ensemble methods	[21]	IEEE TKDE	687.19	165.05	TO	5
M26		[25]	Information Fusion	244.41	31.15	TO	2
M27		[126]	Neurocomputing	5.77	1.38	101.65	0
M28		[126]	Neurocomputing	5.84	1.40	107.27	0
M29		[127]	IEEE TSMCS	1.39	0.57	19.27	0
M30		[127]	IEEE TSMCS	1.39	0.56	19.45	0
M31		[20]	IEEE TKDE	1.16	0.15	29.28	0
M32		[20]	IEEE TKDE	1.19	0.15	31.41	0
M33		[20]	IEEE TKDE	1.21	0.15	32.09	0
M34		[20]	IEEE TKDE	1.19	0.15	30.60	0
M35		[128]	ACM TKDD	221.49	104.50	3051.70	0
M36		[19]	ICML	1.83	0.81	24.31	0
M37		[23]	AAAI	137.82	21.03	2941.91	0
M38		[26]	IJCAI	5.39	2.70	46.25	0
M39	Probabilistic and statistical methods	[29]	ICDM	181.06	19.79	TO	1
M40		[30]	IEEE TPAMI	239.72	29.12	TO	1
M41		[41]	IEEE TPAMI	19.74	1.79	427.71	0
M42		[35]	Pattern Recognition	1.13	0.59	14.12	0
M43		[129]	IEEE TPAMI	22.58	2.33	459.27	0
M44		[130]	ICML	217.13	26.46	TO	1
M45		[32]	IEEE TKDE	20.24	1.25	788.86	0
M46	Weighting-based ensemble methods	[131]	Pattern Recognition Letters	43.02	12.65	680.79	0
M47		[132]	IEEE TNNLS	3.64	1.27	62.45	0
M12		[133]	IEEE TCYB	0.96	0.46	11.23	0
M13		[133]	IEEE TCYB	0.98	0.46	12.02	0
M48		[134]	IEEE TCYB	779.62	1168.31	1266.46	0
M49		[134]	IEEE TCYB	785.96	1178.78	1266.79	0
M50		[134]	IEEE TCYB	776.42	1159.41	1268.24	0
M51		[135]	Information Fusion	1.08	0.46	16.48	0
M52		[136]	IEEE TNNLS	180.10	23.33	3548.82	0
M53		[58]	Pattern Recognition	519.08	76.30	TO	2
M54		[63]	ICDM	19.77	6.65	156.45	0
M55		[63]	ICDM	19.58	6.50	148.20	0
M56		[63]	ICDM	19.59	6.54	142.35	0
M57		[137]	Bioinformatics	0.94	0.54	11.27	0
M58	Uncertainty-based ensemble methods	[138]	Applied Artificial Intelligence	8.23	1.44	324.69	0
M59		[139]	Artificial Intelligence in Medicine	39.67	7.46	400.10	0
M60		[82]	KBS	36.67	3.32	1455.98	0
M61		[92]	Information Sciences	3.01	0.55	42.89	0
M62		[78]	FSS	17.50	1.71	303.63	0
M63		[75]	IEEE TFS	61.07	18.56	942.63	0
M64		[75]	IEEE TFS	60.75	18.13	884.95	0
M65	Representation enhancement methods	[105]	IEEE TNNLS	0.27	0.16	3.76	0
M66		[140]	PR	1.24	1.12	7.85	0
M67		[141]	Bioinformatics	0.26	0.17	2.00	0
M68		[104]	ICME	1.83	1.27	14.97	0
M69		[142]	Information Sciences	1.75	1.25	13.42	0
M70		[98]	IEEE TITS	0.65	0.69	5.71	0
M71	Scalable ensemble methods	[108]	Pattern Recognition	0.38	0.20	7.10	0
M72		[110]	IJAR	0.91	0.27	17.67	0
M73		[143]	Artificial Intelligence Review	5.62	1.46	92.17	0
M74		[113]	Information Fusion	1.42	1.03	6.12	0
M75		[144]	IEEE TKDE	0.42	0.39	1.12	0
M76		[145]	IEEE TKDE	0.48	0.43	1.13	0

TABLE S4
RUNNING TIME RESULTS FOR METHOD VARIANTS M1–M76 ON DATASETS D1–D21.

Method ID	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21
M1	0.68	0.71	0.55	0.60	0.44	0.62	0.64	0.62	0.69	0.55	0.63	0.60	0.60	0.80	0.98	0.62	0.51	0.61	0.66	0.71	0.64
M2	382.04	4.35	4.39	33.10	0.40	0.44	0.53	25.16	3.18	27.49	3.02	4.21	42.00	34.60	29.88	85.12	0.43	6.53	10.12	1804.99	1.00
M3	389.92	3.97	4.30	32.51	0.41	0.44	0.54	25.92	3.19	29.01	2.98	4.36	45.63	33.63	31.82	90.35	0.44	6.41	10.07	1795.01	1.02
M4	1844.46	7.30	15.12	1786.30	0.54	0.62	1.02	93.63	12.78	77.90	7.81	13.26	1719.29	82.08	83.43	1830.20	0.74	18.37	48.63	1774.33	2.31
M5	1755.54	7.57	16.00	1813.70	0.53	0.59	0.99	91.40	12.26	73.85	7.68	14.25	1880.71	86.03	81.00	1769.80	0.73	18.44	53.30	1825.67	2.25
M6	0.82	0.29	0.35	0.42	0.27	5.14	0.31	0.40	0.29	0.45	0.24	0.25	0.47	0.40	0.39	0.73	0.30	0.37	0.29	12.05	0.23
M7	47.96	2.99	2.86	16.19	0.35	0.37	0.42	12.75	1.77	25.12	2.08	3.50	19.59	13.46	13.18	27.19	0.38	6.24	7.13	305.46	0.48
M8	15.72	2.78	2.41	2.46	2.54	2.44	2.08	2.58	2.78	2.89	2.46	2.27	5.24	3.97	3.77	7.05	2.24	2.81	2.92	56.79	2.33
M9	2086.44	64.83	81.68	433.63	1.34	2.19	5.57	421.38	45.00	385.09	39.16	56.15	634.65	427.53	452.99	919.24	2.20	111.34	229.27	TO	14.03
M10	TO	TO	TO	TO	413.93	412.83	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	411.82	TO	TO	TO	TO
M11	25.01	1.25	1.27	7.97	0.52	0.76	0.69	6.28	1.08	9.21	0.94	1.08	5.18	4.74	5.08	15.05	0.62	1.66	2.95	108.11	0.79
M12	0.80	0.31	0.28	0.55	0.38	0.24	0.26	0.41	0.38	0.59	0.36	0.25	0.52	0.55	0.41	0.58	0.32	0.37	0.42	3.43	0.40
M13	0.80	0.33	0.29	0.57	0.37	0.24	0.25	0.41	0.38	0.60	0.36	0.24	0.54	0.55	0.39	0.58	0.34	0.40	0.43	3.54	0.37
M14	25.89	2.32	2.96	6.22	0.48	0.47	0.53	11.72	1.39	3.85	1.17	1.89	19.34	10.14	12.72	16.39	0.45	2.34	4.77	359.90	0.92
M15	26.29	2.22	2.92	6.25	0.50	0.45	0.53	12.48	1.34	3.70	1.23	1.92	19.50	9.94	12.74	16.97	0.42	2.40	4.74	353.32	0.92
M16	175.97	5.72	8.47	48.30	0.36	0.76	0.61	42.98	4.54	41.63	4.36	5.50	56.91	46.24	47.30	105.52	0.43	13.62	21.95	1659.33	1.06
M17	0.14	0.34	0.14	0.14	0.14	0.09	0.16	0.26	0.11	0.14	0.12	0.15	0.30	0.34	0.15	0.12	0.11	0.26	0.23	0.24	0.33
M18	187.49	6.17	4.50	26.40	0.73	0.55	0.62	22.67	2.51	16.97	3.84	3.98	61.04	62.88	35.61	90.76	0.55	13.99	23.44	2882.49	0.94
M19	TO	5.85	6.37	TO	0.86	0.96	1.29	66.65	4.21	84.94	4.36	4.88	TO	188.89	188.97	TO	0.95	14.12	25.44	TO	1.54
M20	12.57	3.04	2.73	5.11	0.19	0.22	0.31	5.33	2.40	5.18	1.40	2.11	4.89	5.07	4.73	9.58	0.33	2.98	3.69	46.31	0.49
M21	TO	23.89	26.06	TO	0.67	0.88	1.74	145.35	17.58	121.60	15.07	21.08	TO	156.15	156.13	TO	0.93	46.48	75.82	TO	4.45
M22	585.80	15.47	17.47	121.47	0.55	0.66	1.55	107.58	11.55	104.73	10.00	14.60	152.35	119.09	121.02	334.03	0.70	32.46	55.28	TO	3.32
M23	TO	46.49	51.35	TO	1.08	2.45	7.02	320.37	35.64	362.89	30.97	45.92	TO	404.24	388.30	TO	2.61	90.66	165.38	TO	12.78
M24	1.77	0.31	0.32	0.68	0.24	0.20	0.21	0.61	0.24	0.65	0.24	0.25	0.82	0.64	0.63	1.11	0.21	0.42	0.53	11.39	0.25
M25	949.70	71.30	61.94	172.03	6.02	6.08	9.01	169.78	46.24	174.64	42.02	56.82	189.90	172.61	175.30	574.60	6.58	89.45	127.16	TO	11.93
M26	179.80	29.92	8.15	34.21	1.72	1.78	1.57	2.83	7.80	4.98	5.03	6.03	128.93	34.16	41.03	81.75	1.75	7.18	81.26	604.08	2.92
M27	4.86	0.51	0.59	1.55	0.22	0.29	0.29	1.44	0.44	1.29	0.39	0.45	1.24	1.59	1.50	2.93	0.35	0.66	1.09	31.73	0.44
M28	4.69	0.55	0.57	1.56	0.22	0.30	0.26	1.42	0.42	1.38	0.38	0.43	1.25	1.53	1.48	2.88	0.36	0.61	1.11	29.59	0.43
M29	1.22	0.43	0.34	0.57	0.29	0.30	0.31	0.58	0.38	0.54	0.33	0.35	0.65	0.61	0.56	0.82	0.29	0.36	0.51	5.37	0.36
M30	1.22	0.40	0.33	0.56	0.28	0.32	0.33	0.62	0.38	0.53	0.37	0.32	0.59	0.59	0.55	0.84	0.29	0.37	0.51	5.53	0.37
M31	0.70	0.14	0.11	2.10	0.12	0.08	0.08	0.12	0.13	0.24	0.10	0.13	1.22	2.04	0.25	1.56	0.11	0.14	0.15	4.90	0.10
M32	0.71	0.13	0.10	2.19	0.11	0.09	0.09	0.12	0.12	0.24	0.11	0.13	1.22	2.06	0.24	1.52	0.10	0.14	0.15	5.22	0.11
M33	0.72	0.13	0.11	2.11	0.11	0.08	0.08	0.12	0.12	0.24	0.11	0.13	1.30	2.05	0.25	1.62	0.10	0.13	0.14	4.94	0.11
M34	0.70	0.13	0.11	2.21	0.11	0.09	0.09	0.12	0.13	0.24	0.10	0.13	1.25	2.05	0.24	1.51	0.10	0.13	0.13	5.34	0.11
M35	178.49	90.37	89.47	103.19	88.78	91.07	84.93	83.92	88.54	73.19	88.07	93.57	108.67	116.68	113.20	153.33	93.08	94.12	97.26	897.31	84.47
M36	1.36	0.70	0.92	0.77	0.52	0.47	0.53	0.65	0.58	0.71	0.48	0.57	0.78	0.74	0.88	0.91	0.55	0.56	0.65	5.84	0.64
M37	33.33	4.92	4.06	14.72	0.42	0.83	0.61	7.99	2.56	10.76	1.42	2.30	37.77	25.83	17.21	26.70	0.53	11.60	13.40	335.29	1.39
M38	7.30	1.23	1.39	3.18	7.27	0.61	0.68	3.03	1.35	3.28	1.18	1.29	2.42	2.29	2.82	5.14	0.73	2.15	1.94	20.29	1.09
M39	110.56	19.05	9.48	3.11	0.22	0.39	0.28	4.18	4.11	9.90	2.31	1.60	206.86	74.42	19.61	24.46	0.44	19.96	49.14	871.60	5.45
M40	155.35	29.02	17.34	8.07	0.31	0.34	0.75	6.97	8.13	14.97	4.02	2.03	317.39	107.23	27.72	46.34	0.81	24.38	83.79	1082.15	9.81
M41	11.02	0.32	0.49	2.54	0.11	0.13	0.14	2.10	0.38	1.46	0.31	0.39	3.32	3.13	2.47	7.56	0.11	0.58	0.69	110.23	0.16
M42	1.00	0.41	0.49	0.44	0.48	0.46	0.58	0.53	0.63	0.48	0.52	0.48	0.60	0.59	0.57	1.04	0.52	0.48	0.54	4.14	0.53
M43	12.48	1.33	0.79	1.86	0.45	0.29	0.54	1.91	0.73	2.35	0.81	0.74	3.61	2.57	2.50	7.88	0.41	1.39	1.75	161.90	0.49
M44	156.58	23.22	3.32	22.09	0.92	4.57	3.09	19.23	2.93	31.53	17.53	25.54	27.38	16.13	15.71	110.09	1.87	36.70	89.29	1956.35	1.53
M45	5.96	0.87	0.56	0.76	0.42	0.37	0.37	0.75	0.62	0.86	0.47	0.48	10.09	3.29	1.31	2.13	0.45	1.29	4.32	54.14	0.43
M46	32.85	4.42	5.10	10.61	2.17	1.30	1.74	9.88	3.82	21.71	3.09	3.74	12.58	10.80	10.85	21.21	1.57	8.38	10.02	211.68	2.08
M47	3.14	0.85	0.79	1.11	0.63	0.63	0.67	1.04	0.90	1.22	0.75	0.78	1.97	1.58	1.32	1.84	0.67	1.04	1.20	16.99	0.72
M48	1157.70	140.71	142.32	1178.58	144.28	139.71	146.69	136.91	141.52	1172.76	135.43	140.72	1204.97	1142.80	1234.97	1221.65	141.81	146.40	138.61	1181.69	141.84
M49	1174.07	136.40	138.74	1195.18	141.39	136.12	135.31	143.04	141.19	1211.30	146.94	139.35	1193.51	1219.96	1221.76	1219.78	144.03	134.75	135.44	1186.38	140.35
M50	1268.24	141.51	137.15	1226.25	133.24	139.06	136.73	140.65	135.92	1215.94	137.13	137.34	1201.52	1237.24	1143.27	1158.57	132.85	136.69	146.09	1231.93	138.75
M51	0.69	0.31	0.56	0.47	0.33	0.41	0.45	0.66	0.31	0.45	0.43	0.29	0.43	0.35	0.36	0.55	0.29	0.53	0.32	3.50	0.29
M52	100.13	6.69	3.59	25.16	0.31	0.33	0.45	12.74	3.04	12.07	2.77	3.31	36.84	45.88	26.47	57.74	0.31	12.82	21.62	1028.63	1.48
M53	737.57	27.54	15.65	75.29	0.83	0.86	1.83	71.03	7.86	65.98	6.36	9.17	142.30	100.45	78.00	426.46	1.14	31.25	91.38	2890.24	4.46
M54	22.26	2.02	1.86	6.60	0.69	0.69	0.80	6.87	1.59	6.71	1.43	1.83	8.03	6.70	7.56	13.95	0.79	2.91	4.24	115.85	1.05
M55	20.95	1.93	1.91	7.13	0.65	0.68	0.81	6.34	1.64	6.58	1.48	1.92	8.35	7.29	7.62	14.02	0.76	2.98	4.42	118.20	1.13
M56	22.21	1.93	1.87	6.71	0.66	0.65	0.86	6.54	1.56	6.53	1.49	1.83	8.19	7.06	7.81	14.41	0.77	2.82	4.45	118.72	1.05
M57	0.77	0.37	0.54	0.46	0.30	0.29	0.30	0.43	0.34	0.61	0.37	0.97	0.48	0.72	0.64	0.58	0.32	0.52	0.46	2.81	0.32
M58	2.85	0.98	0.95	0.90	0.13	0.12	0.15	0.61	0.48	1.46	0.34	0.34	6.09	3.12	1.65	1.73	0.21	1.30	3.04	33.87	0.23
M59	20.56	4																			

TABLE S5
RUNNING TIME RESULTS FOR METHOD VARIANTS M1–M76 ON DATASETS D22–D42.

Method ID	D22	D23	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42
M1	0.54	1.50	0.68	0.64	0.71	1.02	0.83	0.87	0.57	0.57	1.16	0.47	1.10	0.89	0.68	0.70	0.64	0.75	0.56	0.74	0.66
M2	0.57	232.62	2.69	453.40	0.72	0.95	1807.36	50.54	95.41	48.61	1753.23	1.11	1771.43	1642.92	0.56	76.36	105.98	27.58	657.50	30.56	
M3	0.57	240.11	2.79	450.64	0.72	0.97	1792.64	50.25	102.75	46.95	1846.77	1.13	1828.57	1653.84	0.55	78.99	5.31	97.84	28.44	638.30	31.25
M4	1.08	1772.42	6.29	1777.63	0.48	1.09	1802.43	100.93	148.10	1774.33	1809.96	2.78	1748.56	1802.31	1.18	1757.05	8.62	1804.71	89.78	1765.23	84.00
M5	1.11	1827.58	6.37	1822.37	0.46	1.11	1797.57	104.02	144.73	1825.67	1790.04	2.72	1851.44	1797.69	1.28	1842.95	8.20	1795.29	86.15	1834.77	88.34
M6	0.29	0.78	0.33	1.09	0.79	0.36	14.58	0.50	0.52	0.54	2.61	0.36	20.58	9.15	0.37	0.46	0.32	0.56	0.35	1.12	0.46
M7	0.43	38.37	1.69	62.10	0.50	0.53	948.83	15.83	29.77	15.91	159.60	0.50	1404.27	115.39	0.41	16.32	4.86	24.31	12.89	62.64	14.23
M8	2.37	6.23	2.31	4.48	5.39	2.09	276.32	2.94	2.54	2.69	14.65	2.25	164.71	12.23	2.16	11.17	2.16	12.17	12.33	15.22	3.00
M9	7.55	1856.61	30.67	2780.10	1.79	10.27	TO	565.20	1047.03	424.37	TO	11.93	TO	TO	5.93	580.91	102.88	964.80	448.95	TO	513.88
M10	415.76	TO	TO	TO	411.53	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO
M11	0.61	21.33	1.08	28.59	0.64	0.97	213.50	11.10	17.22	4.40	56.16	0.84	306.43	41.48	0.63	12.55	1.61	13.98	5.07	29.03	4.94
M12	0.39	0.65	0.32	0.90	0.46	0.36	8.98	0.47	0.59	0.46	2.05	0.38	11.23	1.43	0.31	0.48	0.41	0.63	0.47	0.96	0.46
M13	0.37	0.68	0.33	0.97	0.45	0.37	9.11	0.47	0.59	0.46	2.06	0.38	12.02	1.36	0.31	0.49	0.38	0.68	0.47	0.93	0.45
M14	0.78	21.88	1.10	63.46	0.42	0.90	1414.84	20.07	16.96	16.61	307.99	0.68	1830.05	162.40	0.59	18.94	1.78	28.16	11.89	95.63	10.01
M15	0.85	23.69	1.11	62.31	0.43	0.89	1292.89	20.24	16.86	15.94	300.94	0.69	1769.95	163.11	0.56	17.65	1.85	26.66	11.57	87.04	9.33
M16	0.93	140.71	3.54	235.76	0.46	1.07	TO	51.46	103.12	54.22	691.20	1.21	TO	456.72	0.85	65.38	13.46	77.46	39.04	261.70	44.29
M17	0.17	0.16	0.29	0.14	0.43	0.14	0.31	0.29	0.27	0.13	0.31	0.12	1.02	0.16	0.15	0.26	0.29	0.57	0.32	0.27	0.12
M18	0.88	176.35	2.80	396.44	1.59	0.82	TO	48.74	62.13	56.14	1564.88	0.77	TO	600.43	0.64	54.04	8.08	170.94	46.44	1014.63	44.89
M19	1.44	TO	3.40	TO	1.23	1.36	TO	163.96	416.66	TO	TO	1.56	TO	TO	0.99	TO	12.00	TO	74.33	TO	170.08
M20	0.48	9.86	0.51	13.13	0.38	0.52	107.45	4.69	8.45	4.41	26.64	0.51	147.74	17.86	0.46	7.50	2.80	9.31	4.96	13.65	4.62
M21	1.96	TO	12.85	TO	0.94	3.45	TO	183.45	347.17	TO	TO	4.43	TO	TO	1.92	TO	41.60	TO	138.38	TO	154.32
M22	1.50	461.95	8.61	788.07	0.61	2.72	TO	138.94	330.07	141.02	1996.41	3.03	TO	1368.52	1.52	178.74	27.67	218.68	106.21	833.47	118.70
M23	7.66	TO	27.92	TO	0.96	10.98	TO	399.99	809.90	TO	TO	12.48	TO	TO	7.28	TO	81.46	TO	310.01	TO	399.27
M24	0.21	1.40	0.31	2.30	0.37	0.28	36.15	0.71	1.37	0.74	6.21	0.35	47.64	4.01	0.45	1.03	0.34	0.99	0.65	2.38	0.66
M25	8.41	837.27	34.49	1163.32	5.69	11.65	TO	197.26	594.04	161.85	2436.55	12.96	TO	1612.27	9.39	191.29	87.13	415.05	161.79	1180.84	176.32
M26	4.25	127.93	3.75	152.04	3.07	2.07	2078.40	173.53	52.67	40.15	2617.58	2.69	TO	316.69	1.76	69.54	12.32	248.86	37.55	146.24	31.60
M27	0.26	4.06	0.43	6.31	0.48	0.45	66.16	1.78	3.11	1.50	14.66	0.32	101.65	9.51	0.27	2.54	0.65	2.50	1.32	6.47	1.40
M28	0.25	3.79	0.41	6.21	0.46	0.43	71.59	1.84	3.03	1.57	15.16	0.30	107.27	9.37	0.25	2.38	0.59	2.48	1.30	6.42	1.43
M29	0.31	1.05	0.33	1.39	0.62	0.33	14.74	0.57	0.79	0.57	3.00	0.31	19.27	2.22	0.30	0.73	0.35	0.75	0.54	1.60	0.59
M30	0.31	1.07	0.34	1.44	0.61	0.31	14.87	0.60	0.83	0.62	3.27	0.29	19.45	2.22	0.29	0.74	0.37	0.78	0.54	1.53	0.54
M31	0.09	0.69	0.11	0.33	0.12	0.09	29.28	0.14	0.12	1.13	1.45	0.10	17.40	0.85	0.09	1.05	0.10	0.59	0.15	0.98	0.15
M32	0.09	0.65	0.11	0.33	0.12	0.09	31.41	0.14	0.12	1.15	1.39	0.10	17.80	0.86	0.09	1.11	0.11	0.64	0.15	1.04	0.15
M33	0.09	0.69	0.11	0.33	0.12	0.10	32.09	0.14	0.12	1.18	1.32	0.10	18.58	0.81	0.08	1.06	0.12	0.62	0.15	1.05	0.16
M34	0.09	0.68	0.12	0.35	0.12	0.09	30.60	0.15	0.11	1.17	1.41	0.10	18.61	0.87	0.08	1.04	0.11	0.64	0.16	1.05	0.15
M35	92.76	168.17	89.30	213.31	92.02	88.39	1780.50	114.58	134.07	111.48	374.37	87.73	3051.70	348.98	91.63	143.87	82.74	156.62	105.81	268.57	106.18
M36	0.53	1.19	0.87	1.50	0.51	0.48	15.44	0.95	0.91	0.82	3.18	0.54	24.31	2.08	0.47	0.80	0.58	1.43	0.67	1.62	1.70
M37	1.02	94.42	3.16	113.29	0.95	8.31	1700.38	26.81	72.11	32.39	516.48	1.42	2941.91	108.81	1.15	21.54	5.98	52.86	23.42	177.92	29.99
M38	0.72	4.71	0.82	7.09	1.07	0.68	38.30	2.70	5.34	2.48	12.34	0.90	46.25	8.24	0.73	4.88	1.74	3.39	2.58	6.61	3.20
M39	6.51	167.28	4.64	59.06	0.40	0.47	581.43	29.52	11.84	24.20	1790.96	1.97	TO	335.26	1.88	21.90	2.13	1297.98	42.85	525.67	30.02
M40	8.58	352.93	11.10	115.60	0.65	0.87	796.73	50.14	18.81	43.15	2369.81	3.55	TO	494.76	3.89	29.22	3.67	1365.24	71.18	741.43	49.49
M41	0.11	9.34	0.29	19.85	0.15	0.14	320.43	2.41	5.91	2.69	44.24	0.18	427.71	20.07	0.11	4.85	0.68	0.39	2.16	15.78	2.03
M42	0.62	1.07	0.63	0.85	1.09	0.47	14.12	0.59	0.61	0.48	1.62	0.43	9.53	1.40	0.59	1.07	0.62	0.91	0.58	1.55	0.51
M43	0.45	10.17	1.47	13.97	0.33	0.71	221.86	3.54	8.98	2.66	77.22	0.65	459.27	23.22	0.69	3.38	1.21	14.50	2.32	18.33	2.87
M44	1.00	248.32	15.99	99.69	1.65	2.33	2973.49	42.81	91.94	67.53	222.69	4.08	TO	1009.89	5.20	207.59	12.50	234.33	185.97	87.72	19.38
M45	0.63	14.96	0.45	6.56	0.81	0.36	36.34	2.37	1.74	1.89	231.22	0.48	788.86	24.19	0.48	0.98	0.58	81.13	2.14	28.97	2.06
M46	1.67	35.83	3.06	43.33	1.70	2.01	564.45	12.06	21.26	12.76	101.41	1.96	680.79	75.65	1.96	15.47	6.80	17.89	19.43	53.35	10.32
M47	0.70	3.08	0.84	2.64	1.63	0.75	43.64	1.35	1.63	1.52	7.92	0.75	62.45	5.66	0.68	1.78	0.86	4.18	1.33	5.67	1.30
M48	140.05	1185.20	144.37	1211.33	144.10	136.72	1238.65	1222.62	1203.88	1157.42	1219.10	143.09	1187.38	1166.74	140.31	1202.90	1210.32	1219.77	1202.28	1172.75	1266.46
M49	140.79	1238.78	137.89	1233.31	137.90	147.37	1229.85	1171.30	1198.52	1231.40	1241.40	138.53	1219.19	1178.56	140.20	1191.18	1178.99	1219.98	1191.36	1193.63	1157.80
M50	136.09	1176.03	137.16	1155.36	137.60	134.52	1131.50	1206.08	1197.60	1211.18	1139.50	138.26	1193.43	1254.70	138.82	1205.92	1210.69	1160.25	1206.36	1233.62	1175.75
M51	0.45	0.69	0.37	1.09	0.48	0.40	11.98	0.65	0.50	0.52	2.14	0.38	16.48	1.51	0.25	0.82	0.45	0.56	0.34	0.87	0.41
M52	1.05	79.19	2.05	126.98	0.63	0.66	2774.92	31.42	53.20	30.36	790.11	0.91	3548.82	281.06	0.45	39.65	5.89	63.29	24.11	157.33	25.99
M53	3.20	576.01	6.34	855.32	0.86	2.90	TO	106.39	417.41	82.53	2395.13	3.92	TO	1227.22	2.36	112.13	18.56	636.93	87.64	1032.39	88.47
M54	0.78	18.94	1.52	26.97	0.81	1.08	68.97	8.12	13.37	8.01	77.52	0.89	89.35	56.27	0.87	10.20	2.48	11.74	6.23	30.03	6.98
M55	0.83	19.13	1.39	26.43	0.84	1.03	68.66	8.00	13.59	8.20	76.86	0.92	85.77	57.06	0.86	10.48	2.49	11.93	6.49	31.79	6.71
M56	0.84	19.39	1.42	26.20	0.89	1.11	69.14	8.33	13.54	7.85	77.79	0.94	89.62	60.26	0.91	10.16	2.59	11.74	6.25	29.93	6.64
M57	0.34	0.80	0.32	0.93	0.47	0.31	7.73	1.10	0.76	0.58	1.66	0.36	11.27	1.96	0.31	0.52	0.40	0.61	0.67	0.87	0.43
M58	0.17	5.97	0.34																		

TABLE S6
RUNNING TIME RESULTS FOR METHOD VARIANTS M1–M76 ON DATASETS D43–D64.

Method ID	D43	D44	D45	D46	D47	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	D64
M1	0.88	0.65	0.59	0.57	0.60	0.65	0.95	0.55	0.50	0.55	0.54	0.62	5.58	0.61	1.09	0.82	0.75	0.64	0.70	0.48	0.70	0.84
M2	934.25	953.62	3.93	0.51	753.02	27.15	1785.13	55.23	0.47	11.79	263.90	5.23	1774.32	0.56	79.17	13.76	780.67	901.54	909.93	0.52	22.54	1686.33
M3	932.36	940.49	4.09	0.51	750.59	26.07	1814.87	54.72	0.47	11.90	265.63	5.06	1825.68	0.55	78.17	13.22	824.79	901.07	886.36	0.51	24.69	1686.14
M4	1835.86	1793.53	4.50	0.63	1790.73	62.16	1786.85	1785.18	0.81	49.46	1869.73	23.84	1755.63	1.19	96.06	60.42	1846.08	1719.46	1825.96	1.25	63.83	1770.61
M5	1764.14	1806.47	4.65	0.66	1809.27	61.83	1813.15	1814.82	0.79	51.73	1730.27	22.69	1844.37	1.27	95.76	58.25	1753.92	1880.54	1774.04	1.23	61.95	1829.39
M6	1.40	1.36	0.28	0.31	1.38	0.55	5.13	0.40	0.31	0.33	0.73	0.26	3.30	0.23	0.59	0.40	4.64	1.27	1.27	0.24	0.54	1.81
M7	81.34	80.71	1.25	0.36	78.86	16.21	358.99	17.72	0.36	9.65	35.07	4.26	289.95	0.41	23.72	7.60	85.98	78.15	81.83	0.45	11.98	112.20
M8	8.81	12.30	3.12	2.33	9.16	7.49	15.98	3.21	2.48	2.50	16.02	2.17	2.53	2.33	2.49	3.62	3.55	5.75	3.21	2.13	2.39	12.88
M9	TO	TO	25.01	6.53	TO	339.62	TO	507.62	1.89	224.46	1623.01	122.15	TO	6.72	704.36	270.61	TO	TO	TO	5.17	332.26	TO
M10	TO	TO	TO	412.60	TO	TO	TO	TO	415.05	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO
M11	26.38	36.02	1.05	0.65	35.71	7.20	69.52	11.66	0.68	1.92	22.23	1.50	87.51	0.61	7.78	3.58	34.32	37.09	37.88	0.58	7.28	47.92
M12	1.32	1.24	0.28	0.28	1.18	0.56	3.71	0.48	0.38	0.41	0.70	0.30	2.97	0.23	0.65	0.38	1.35	1.24	1.13	0.32	0.45	1.49
M13	1.30	1.29	0.28	0.29	1.12	0.55	3.99	0.49	0.40	0.41	0.68	0.30	3.03	0.24	0.65	0.36	1.27	1.35	1.19	0.31	0.46	1.52
M14	53.12	168.06	2.14	0.71	38.19	3.83	410.41	14.97	0.45	2.12	35.19	2.15	204.50	0.51	16.50	7.36	53.15	111.94	58.88	0.54	12.99	203.09
M15	50.78	155.24	2.14	0.66	39.65	4.00	408.18	15.89	0.43	2.25	35.64	2.03	199.77	0.52	15.89	7.45	52.17	109.69	58.54	0.50	12.65	194.38
M16	341.83	337.99	2.55	0.47	345.26	41.08	1792.69	61.57	0.46	25.36	141.30	10.39	1518.20	0.82	65.82	28.05	315.16	329.79	326.27	1.02	36.94	437.87
M17	0.15	0.17	0.34	0.53	0.28	0.10	0.23	0.11	0.20	0.14	0.12	0.39	0.34	0.14	0.28	0.45	0.31	0.23	0.24	0.27	0.29	0.17
M18	544.11	472.12	3.35	0.82	429.62	21.72	TO	38.59	0.57	7.99	98.74	8.94	2096.53	0.62	39.23	47.59	365.55	536.10	368.05	0.78	27.40	676.84
M19	TO	TO	2.29	1.13	TO	70.11	TO	TO	0.76	28.09	TO	11.12	TO	0.98	231.35	53.64	TO	TO	TO	0.93	68.13	TO
M20	12.59	16.11	0.66	0.35	17.43	4.91	33.22	6.36	0.24	3.13	11.17	2.80	43.77	0.36	6.26	4.23	16.68	17.06	17.42	0.46	4.69	21.26
M21	TO	TO	8.43	0.81	TO	118.98	TO	TO	0.91	60.41	TO	40.02	TO	1.91	230.74	92.64	TO	TO	TO	1.68	132.18	TO
M22	1046.03	1057.32	5.61	0.66	99.05	1015.30	TO	153.27	0.68	55.39	466.05	27.13	3557.75	1.57	172.78	64.89	995.60	984.78	994.47	1.36	95.03	1360.99
M23	TO	TO	17.69	2.25	TO	293.36	TO	TO	2.52	196.81	TO	78.96	TO	7.56	498.27	198.88	TO	TO	TO	6.69	291.07	TO
M24	3.18	3.20	0.22	0.23	2.91	0.69	12.39	0.75	0.19	0.47	1.48	0.45	9.12	0.22	0.84	0.47	3.12	2.72	2.74	0.22	0.66	4.06
M25	1407.74	1621.09	20.78	4.74	1432.20	165.55	3476.70	200.48	6.42	109.59	716.40	83.18	TO	10.21	225.58	143.48	1604.31	1373.31	1537.29	8.73	156.49	1876.22
M26	150.29	177.96	7.22	3.64	586.94	23.32	1084.47	41.51	1.96	5.55	97.78	10.72	464.19	2.10	30.42	278.74	401.85	180.62	158.16	1.88	91.08	329.88
M27	5.82	7.93	0.47	0.36	7.56	1.33	24.00	2.06	0.31	0.72	3.97	2.82	22.59	0.28	2.04	1.34	7.87	7.76	7.92	0.40	1.62	11.15
M28	6.10	8.15	0.46	0.37	7.91	1.23	22.46	1.90	0.32	0.70	3.94	3.02	23.15	0.29	2.01	1.28	7.52	7.41	7.52	0.38	1.62	10.95
M29	1.66	1.80	0.35	0.30	1.85	0.52	5.70	0.68	0.30	0.58	1.02	0.36	4.39	0.29	0.68	0.52	1.62	1.70	1.70	0.30	0.56	2.27
M30	1.82	1.86	0.32	0.29	1.82	0.55	5.31	0.67	0.28	0.59	1.03	0.35	4.40	0.30	0.67	0.50	1.68	1.60	1.69	0.31	0.53	2.20
M31	0.57	0.91	0.11	0.09	0.94	0.16	1.10	0.33	0.11	0.12	0.68	0.10	0.21	0.08	0.10	0.22	0.28	0.54	0.18	0.09	0.11	0.97
M32	0.59	0.97	0.11	0.08	0.87	0.17	1.02	0.33	0.12	0.12	0.71	0.10	0.21	0.08	0.11	0.22	0.27	0.54	0.19	0.09	0.11	1.02
M33	0.57	0.96	0.11	0.09	0.95	0.16	1.11	0.32	0.11	0.12	0.70	0.10	0.21	0.08	0.11	0.22	0.27	0.56	0.18	0.09	0.12	0.96
M34	0.59	0.90	0.11	0.09	0.95	0.17	1.10	0.32	0.12	0.13	0.64	0.11	0.20	0.09	0.11	0.21	0.28	0.56	0.19	0.09	0.11	0.95
M35	212.52	294.82	84.91	91.36	225.19	102.03	509.48	123.74	91.11	88.76	181.78	80.28	550.28	90.28	120.61	106.14	231.95	273.21	244.35	86.53	106.80	320.37
M36	1.75	1.83	0.61	0.57	1.76	0.61	13.00	0.91	0.60	0.63	1.08	0.58	5.22	5.11	1.02	0.97	5.98	1.67	1.77	0.48	0.86	2.15
M37	266.01	240.84	2.36	0.76	257.45	38.13	1096.22	41.08	0.52	21.42	29.74	2.46	985.01	1.10	40.98	15.61	221.30	56.22	125.35	0.83	20.64	92.56
M38	6.89	8.30	0.76	0.70	10.65	3.16	17.79	3.87	0.62	1.91	6.64	1.60	18.72	0.70	3.74	2.04	9.87	7.86	9.13	0.63	2.14	11.23
M39	360.89	541.73	49.72	1.43	304.35	10.25	766.17	7.14	0.23	4.21	60.77	1.39	102.36	0.54	6.48	37.95	114.13	140.91	80.94	0.52	20.56	434.81
M40	588.29	814.16	79.12	3.52	424.35	14.55	1316.76	14.95	0.58	6.97	94.76	2.48	151.14	1.05	6.89	73.77	212.33	275.81	123.60	0.85	32.71	553.67
M41	34.08	30.87	0.08	0.09	27.42	1.80	110.64	3.91	0.13	1.00	10.08	0.73	50.69	0.10	3.61	1.53	17.94	24.44	21.46	0.11	1.71	26.68
M42	1.08	1.40	0.48	0.46	1.18	0.46	1.68	0.52	0.59	0.44	1.25	0.42	0.70	0.59	0.55	0.55	0.79	0.96	0.66	0.46	0.46	1.66
M43	35.30	22.52	0.74	0.41	18.37	2.27	147.89	3.06	0.30	1.38	9.98	1.19	213.10	0.46	2.42	1.70	18.15	17.84	16.61	0.32	2.64	22.16
M44	105.01	162.73	16.12	2.48	246.29	47.64	417.62	24.08	0.84	39.06	148.78	5.39	580.38	4.71	308.84	22.21	130.19	245.80	164.87	6.80	51.94	533.42
M45	22.47	36.53	1.21	0.42	17.78	1.03	64.72	1.13	0.37	0.80	2.40	0.54	15.01	0.41	0.72	2.56	7.88	11.02	10.26	0.39	1.53	21.60
M46	49.14	59.45	2.43	1.54	51.35	22.39	170.42	12.84	1.39	12.88	28.89	6.57	168.11	1.61	13.50	16.21	49.75	56.50	52.00	2.00	18.63	77.88
M47	3.63	4.90	0.93	0.69	4.90	1.08	9.69	1.32	0.65	0.88	2.46	0.80	8.80	0.63	1.23	1.45	3.33	3.68	3.24	0.69	1.21	6.20
M48	1194.46	1225.31	141.94	141.38	1185.10	1187.44	1204.50	1172.98	135.63	137.15	1212.47	1201.41	1206.18	135.71	1227.54	1210.65	1143.88	1210.85	1196.97	144.25	1237.30	1166.68
M49	1222.27	1159.17	136.44	143.09	1234.77	1219.86	1186.53	1231.21	143.48	136.22	1199.50	1198.35	1254.59	140.20	1237.39	1166.30	1245.26	1192.04	1170.12	135.93	1150.52	1266.79
M50	1183.27	1215.53	137.46	131.17	1180.12	1192.70	1208.97	1195.81	137.88	141.10	1188.03	1200.24	1139.24	139.78	1135.07	1223.05	1210.86	1197.12	1232.91	137.28	1212.18	1166.52
M51	1.28	1.15	0.40	0.27	1.21	0.34	4.23	0.44	0.25	0.53	0.80	0.45	2.68	0.43	0.44	0.57	1.13	1.32	1.04	0.32	0.43	1.67
M52	187.02	185.03	2.42	0.71	305.88	14.04	923.71	34.75	0.39	7.84	93.19	5.38	757.87	0.50	36.25	15.86	186.14	191.74	182.08	0.44	22.74	271.91
M53	977.66	1186.22	33.10	1.87	1012.67	62.38	2290.32	99.92	1.07	31.26	468.47	17.44	2644.07	2.15	107.26	77.31	1121.93	992.38	1023.67	1.65	73.15	1421.72
M54	39.38	42.29	1.04	0.67	35.47	6.33	121.27	9.52	0.72	3.83	21.46	2.66	156.45	0.86	8.95	4.88	33.36	35.32	32.04	0.72	6.35	48.78
M55	42.40	41.73	1.08	0.67	34.59	6.37	116.02	9.47	0.68	3.80	20.97	2.53										

TABLE S7
RUNNING TIME RESULTS FOR METHOD VARIANTS M1–M76 ON DATASETS D65–D86.

Method ID	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86
M1	1.52	0.67	0.71	0.53	0.60	0.57	0.81	0.62	0.54	0.70	0.55	0.74	0.94	0.71	0.74	1.37	0.59	0.72	0.65	0.56	0.61	0.74
M2	462.95	1125.62	1770.45	0.50	436.78	3.31	1806.37	4.65	0.50	0.58	567.09	639.33	3.24	8.67	895.28	2.03	0.64	1415.00	22.63	0.50	11.76	0.42
M3	486.66	1085.10	1829.55	0.48	443.59	3.27	1793.63	4.71	0.49	0.58	579.75	596.61	3.13	8.83	891.85	2.02	0.65	1431.44	24.17	0.52	11.23	0.42
M4	1790.05	1839.73	1790.63	1.10	1845.77	9.73	1785.96	11.79	0.73	1.27	1831.48	1807.13	7.83	24.88	1785.88	2.45	0.74	1843.52	80.82	1.06	30.53	0.33
M5	1809.95	1760.27	1809.37	1.15	1754.23	9.16	1814.04	11.74	0.80	1.31	1768.52	1792.87	7.99	24.17	1814.12	2.57	0.77	1756.48	80.30	1.00	30.25	0.35
M6	0.92	2.36	2.22	0.42	1.09	0.36	4.84	0.33	0.49	0.36	1.17	1.32	0.27	0.72	1.22	0.24	0.23	1.75	0.50	0.39	0.36	0.31
M7	57.55	77.24	151.60	0.38	63.07	1.76	321.64	3.12	0.53	0.47	49.17	82.34	2.09	6.98	82.62	1.57	0.54	102.53	12.03	0.40	6.70	0.34
M8	5.76	7.03	2.67	2.43	2.70	2.57	88.33	2.58	2.31	2.20	45.60	48.26	2.64	2.61	3.21	2.30	2.25	21.74	4.36	2.40	3.98	2.30
M9	2273.03	TO	TO	4.75	2421.16	47.40	TO	77.31	3.05	7.15	2916.28	3275.17	38.60	166.80	TO	23.04	4.99	TO	439.41	4.53	193.79	2.11
M10	TO	TO	TO	413.45	TO	TO	TO	TO	414.46	TO	TO	TO	TO	TO	TO	TO	TO	TO	415.32	TO	414.09	
M11	26.54	25.00	57.07	0.77	27.35	1.60	97.02	1.32	0.65	0.62	31.67	33.42	0.92	2.40	34.23	0.93	0.64	46.69	4.25	0.63	2.78	1.00
M12	0.86	1.22	1.87	0.36	0.97	0.47	3.30	0.28	0.36	0.27	0.91	0.86	0.30	0.38	1.20	0.28	0.23	1.32	0.51	0.36	0.39	0.37
M13	0.89	1.19	2.04	0.33	0.97	0.45	3.27	0.26	0.37	0.30	0.91	0.94	0.30	0.38	1.16	0.31	0.23	1.43	0.50	0.37	0.40	0.34
M14	84.77	53.99	54.61	0.51	25.62	2.08	308.01	1.49	0.57	0.83	79.61	76.28	1.22	4.68	57.46	0.88	0.49	166.77	10.40	0.60	5.26	0.39
M15	93.31	50.14	51.17	0.49	24.78	2.11	313.65	1.41	0.55	0.85	76.39	71.97	1.24	5.06	57.13	0.88	0.52	165.87	9.93	0.61	5.17	0.40
M16	207.57	316.53	605.83	0.58	198.46	4.10	1668.83	8.79	0.62	0.71	258.84	255.27	4.41	20.37	314.67	2.69	1.02	469.57	37.50	0.56	25.59	0.46
M17	0.20	0.18	0.11	0.22	0.13	0.16	0.36	0.14	0.09	0.32	0.15	0.21	0.29	0.32	0.24	0.21	0.22	0.20	0.29	0.16	0.35	0.23
M18	226.48	906.98	680.14	0.77	220.59	3.78	TO	7.13	0.63	0.92	290.88	402.96	3.37	9.90	481.30	1.83	0.73	465.59	29.78	0.86	40.44	0.58
M19	TO	TO	TO	0.83	TO	3.20	TO	10.72	0.88	1.09	TO	TO	4.19	32.43	TO	2.59	0.90	TO	76.01	0.96	44.85	0.89
M20	13.00	14.05	23.06	0.29	13.15	2.39	46.32	2.53	0.41	0.40	14.54	13.83	1.48	3.45	16.90	0.45	0.35	22.26	4.02	0.65	3.49	0.28
M21	TO	TO	TO	1.52	TO	15.50	TO	30.30	1.10	1.95	TO	TO	14.38	65.26	TO	9.59	1.57	TO	137.65	1.48	78.07	0.84
M22	685.03	1012.64	1725.79	1.24	705.22	9.95	TO	20.33	0.87	1.59	874.23	836.97	9.76	49.50	991.73	6.48	1.22	1362.28	106.18	1.20	57.16	0.72
M23	TO	TO	TO	5.81	TO	30.36	TO	59.81	3.21	7.70	TO	TO	30.41	143.96	TO	20.12	5.90	TO	310.23	5.61	173.29	2.28
M24	2.14	2.83	5.20	0.21	2.19	0.37	11.98	0.40	0.21	0.22	2.71	2.45	0.23	0.44	2.74	0.24	0.22	4.03	0.60	0.21	0.60	0.25
M25	994.56	1310.98	2225.84	8.39	1051.20	61.09	TO	63.60	6.86	9.09	1403.19	1245.30	42.35	117.93	1556.45	23.84	9.63	2077.32	164.54	7.72	122.81	4.87
M26	128.90	997.14	303.71	1.76	11.89	24.36	TO	7.67	4.21	2.19	191.56	224.83	4.94	16.49	164.35	1.93	2.30	164.91	30.69	2.35	11.58	1.88
M27	5.63	5.78	13.82	0.32	5.28	0.78	26.05	0.52	0.24	0.24	6.72	7.06	0.42	0.80	7.54	0.33	0.28	11.30	1.36	0.26	0.91	0.32
M28	5.57	6.13	13.14	0.31	5.07	0.76	27.21	0.52	0.23	0.25	6.56	6.70	0.42	0.77	7.36	0.31	0.29	10.79	1.33	0.26	0.90	0.31
M29	1.34	1.69	2.78	0.29	1.21	0.50	5.61	0.35	0.30	0.32	1.52	1.64	0.34	0.48	1.59	0.31	0.65	2.23	0.57	0.30	0.46	0.28
M30	1.30	1.71	2.79	0.28	1.26	0.50	5.40	0.34	0.32	0.29	1.56	1.54	0.34	0.46	1.67	0.33	0.66	2.36	0.55	0.30	0.45	0.29
M31	0.55	0.67	0.19	0.09	0.13	0.10	7.16	0.12	0.11	0.08	3.99	4.34	0.10	0.12	0.20	0.13	0.12	1.88	0.30	0.09	0.25	0.09
M32	0.56	0.67	0.19	0.09	0.14	0.11	6.92	0.11	0.11	0.08	4.25	4.24	0.11	0.12	0.19	0.12	0.12	1.87	0.31	0.09	0.24	0.08
M33	0.54	0.66	0.19	0.09	0.13	0.10	7.22	0.11	0.11	0.09	4.31	4.01	0.11	0.12	0.20	0.16	0.12	1.84	0.30	0.09	0.24	0.09
M34	0.51	0.67	0.18	0.09	0.13	0.11	7.07	0.12	0.11	0.08	4.05	3.97	0.10	0.12	0.20	0.18	0.12	1.89	0.32	0.09	0.24	0.09
M35	234.94	189.21	366.44	92.34	82.95	87.37	894.30	80.20	91.03	91.02	294.33	374.53	86.67	95.32	240.05	85.33	90.98	350.97	107.89	91.00	97.34	93.30
M36	1.27	1.73	2.61	0.66	1.40	0.53	5.80	0.59	0.58	0.55	1.51	1.41	0.64	0.70	1.59	0.52	0.54	2.20	0.91	0.55	0.67	0.53
M37	132.88	140.92	310.72	0.71	44.42	3.45	558.24	5.87	0.84	0.98	150.34	118.13	1.39	9.94	187.71	2.30	0.99	97.43	21.89	0.94	17.33	0.84
M38	7.87	4.97	13.88	0.75	8.10	1.17	14.95	1.54	0.65	0.71	10.11	12.83	1.16	2.14	9.20	0.82	0.71	10.61	2.69	0.76	2.75	0.63
M39	66.79	409.61	62.13	0.93	19.57	16.55	1351.41	8.27	1.44	3.24	158.78	171.05	2.39	4.70	76.81	1.93	0.46	100.96	25.18	4.92	62.63	0.68
M40	131.39	735.45	88.44	1.80	38.91	28.50	1760.75	14.26	2.95	6.46	242.40	247.33	3.99	6.48	120.23	3.86	0.77	158.66	40.39	11.25	98.97	2.30
M41	15.87	13.64	49.76	0.12	16.05	0.39	89.75	0.40	0.11	0.11	24.86	25.61	0.29	1.16	20.97	0.21	0.10	38.99	1.77	0.10	1.44	0.10
M42	0.90	1.03	0.60	0.61	0.55	0.48	4.55	0.53	0.49	0.48	3.43	3.58	0.65	0.48	0.64	0.52	0.51	2.38	0.79	0.45	0.65	0.44
M43	13.47	20.55	26.73	0.35	12.81	0.91	164.56	1.27	0.63	0.48	15.71	18.86	0.69	1.37	17.08	0.54	0.57	26.70	2.50	0.50	2.15	0.31
M44	155.43	820.64	794.64	0.91	171.89	21.88	553.12	16.00	0.77	0.40	321.93	428.13	17.58	16.89	165.79	4.66	2.74	179.33	17.40	2.21	11.51	0.85
M45	5.71	30.15	8.04	0.43	5.59	0.79	107.71	0.59	0.42	0.53	10.98	10.42	0.39	0.76	7.90	0.48	0.42	6.25	1.20	0.50	2.84	0.43
M46	38.68	50.81	94.81	1.74	35.07	3.69	244.47	6.05	1.63	1.84	44.67	45.09	3.17	10.29	51.64	2.86	1.63	78.23	18.68	1.96	12.72	1.49
M47	2.73	3.83	5.16	0.66	2.41	0.99	18.45	0.90	0.67	0.73	3.70	4.02	0.81	0.93	3.31	0.73	0.69	5.65	1.22	0.71	1.38	0.71
M48	1158.72	1187.92	1169.88	134.07	1250.73	137.00	1153.77	1198.70	139.76	140.60	1151.76	1219.17	138.88	1205.83	1205.27	133.45	138.12	1225.11	1195.68	143.72	1215.76	135.91
M49	1242.89	1209.18	1246.17	138.50	1194.26	139.04	1258.75	1206.85	144.89	144.67	1192.26	1217.37	139.82	1172.65	1232.69	137.27	143.99	1223.26	1206.17	136.66	1198.83	143.28
M50	1198.40	1202.90	1183.95	145.64	1155.00	141.40	1187.49	1194.44	134.27	134.39	1255.99	1163.46	137.06	1221.52	1162.04	146.33	133.40	1151.63	1198.15	140.56	1185.41	135.78
M51	0.83	1.16	1.85	0.30	0.76	0.28	4.17	0.33	0.37	0.25	0.96	0.94	0.32	0.39	1.21	0.26	0.31	1.45	0.53	0.36	0.48	0.29
M52	117.85	323.66	359.46	0.91	58.87	4.25	917.39	4.65	0.35	1.12	171.77	147.94	2.42	11.15	179.73	1.35	0.48	270.79	23.35	0.77	23.31	0.39
M53	788.28	889.78	1461.66	1.50	787.71	29.70	2952.08	11.53	1.90	3.37	928.78	959.53	6.24	34.15	1055.00	4.64	2.04	1299.46	73.16	4.23	55.36	1.42
M54	26.66	31.65	57.01	0.80	24.24	1.84	116.42	2.00	0.80	0.92	32.96	34.35	1.45	3.65	32.67	1.23	0.89	50.21	6.87	0.75	4.15	0.68
M55	25.36	31.85	58.81	0.85	24.78	1.71	112.23	2.10	0.76	0.95	32.18	32.19	1.49	3.77	35.07	1.20	0.83	50.21	6.50	0.81	4.09	0.70
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